补充材料

来源隔离评估揭示文献催化机器学习的泛化缺口：少标签适配、输入表征与公开研究复现审计

A部分 原始协议与审计补充（4页）

数据构建、特征、预算、原始对照、分配敏感性及公开研究复现细节。页码以A开头。

B部分 扩展方法与完整结果（45页）

额外来源、强本地与残差/混合对照、全部选样矩阵、学习协方差与视图门控、表征交叉、核及风险消融、身份分离和重复划分。页码、图表及公式以B开头。

两部分共同对应同一篇主文；不同分析层采用各自归档的匹配对照，统计口径与来源总体保持分开。

Supplement A: Core protocol and audits
Source-Disjoint Evaluation Reveals Hidden Generalization Gaps in Literature-Derived Catalysis Machine Learning

Zhang Junyu & Li Wenjie

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1 S1 Dataset and provenance details

Methanol conversion/selectivity corpus (1,245 paired records, 82 sources, 80 leakage groups). Rebuilt row-by-row from TheMeCat v1 (`TheMeCat_v1.csv`, Zenodo record 17085762 v1.0, CC BY 4.0, 1,124 rows, SHA-256 46000f00…) and the Communications Engineering methanol synthesis file (`commseng_datameoh.csv`, 500 rows, SHA-256 0a81730a…). Inclusion: both targets reported within $[0, 100]$; temperature, pressure, H_2/CO_2 positive and reported; GHSV/BET allowed missing (imputed within training folds); no performance-based filtering; genuine zeros retained; full-field duplicates kept once. Exclusions (row-level log): -319 no paired target, -26 one target missing, -30 unresolved attribution/invalid

DOI, -1 duplicate. Twenty records whose BET had been reconstructed as 10^{LN} were recomputed from the stored natural-log field as $\exp(\mathrm{LN})$, logged.

STY corpus (1,230 records, 109 sources). Derived from the Suvarna et al. public dataset (CC BY 4.0); 20 DOIs overlap the conversion corpus (declared; treated as a same-reaction different-target task). Four rows dropped for invalid target or missing source mapping (logged).

OCM corpus (1,794 records, 408 sources, 405 leakage groups). From the Schmack et al. meta-analysis supplement (CC BY 4.0): 1,802 raw catalyst records over 412 publication numbers; entries with missing/ambiguous references removed with a row-level log; three pairs of identical-citation sources merged. The published corrected dataset is used together with the original paper's processing description; it is not claimed to be a raw experimental census.

2 S2 Complete feature definitions

Methanol/STY operating features: temperature (K), total pressure (bar), H_2/CO_2 ratio (—), GHSV ($\mathrm{L\,g^{-1}h^{-1}}$; STY corpus: $\mathrm{cm^3\,h^{-1}g^{-1}}$ as published), BET surface area ($\mathrm{m^2g^{-1}}$; methanol corpus only), catalyst amount (STY corpus). **OCM F0:** temperature (K), CH_4 partial pressure (bar), O_2 partial pressure (bar), CH_4/O_2 ratio (—), total pressure (bar), contact time (s). **OCM F1 adds:** up to four cation slots and two support slots with species name, "listed" indicator, known mol-fraction sum (mol$\%/100$), missing-amount count, and an explicit unseen-species channel. Feature vocabulary (species names) is fitted on the current training partition only; sodium is never treated as a missing marker; unlisted species are "unreported", not proven absent. Response variables and their derivatives are excluded from all feature sets.

3 S3 Model hyperparameters

Base-model grid (inner source-grouped 3-fold, inner equal-weight MAE): ridge $\alpha \in \{0.1, 1, 10, 100\}$; gradient-boosted trees $n \in \{100, 300\} \times$depth $\in \{2, 3\}$ (learning rate 0.05, subsample 0.8, min-samples-split 3, seed 42) plus $n = 500$, depth 8. **Calibration hyper-parameters** (penalty λ for affine, α for local ridge, temperature ridge): selected on training-source pseudo-tasks (ten pseudo-repeats per training source, five labels each). **Case-study models:** author configurations reproduced as published (Suvarna: XGB colsample 0.5, eta 0.15, depth 4, n=1000, subsample 1, seed 42; RF depth 40, sqrt, min-split 2, n=500; GBT lr 0.05, depth 4, n=300, subsample 0.5. Xu: Optuna 100-trial tuning with record-level cv=5, author's search spaces).

4 S4 All-k learning curves

Source-equal MAE at $k = 1, 2, 3, 5, 10$ (main cohort, random selector) for the eight display methods on all four targets, with 95% leakage-group bootstrap intervals, are provided in the machine-readable registry (`FINAL_NUMBER_REGISTRY.csv`, `MASTER_RESULTS.json`) and summarized in the main-text Table 3 ($k = 5$) and Table 4 ($k = 1, 5, 10$ paired contrasts). The stacked R^2 is the predeclared primary caliber; per-row-mean and per-seed calibers are archived alongside.

5 S5 Low-budget small-source panel

Sources whose calibration pool holds 3–9 records (methanol 22, OCM 45, STY 43) are reported only at $k \leq 3$ and never mixed into main-cohort numbers. At $k = 3$ the mean offset reduces base MAE on this panel, mirroring the main-cohort picture; the corrected $k = 3$-only panel supersedes the earlier version that mixed pools.

6 S6 Full paired contrasts

Per-source paired contrasts (mean MAE difference, improved/harmed counts, bootstrap CIs) for all method pairs at $k = 1, 5, 10$ and all four targets are in `MASTER_RESULTS.json` (block `paired_comparisons`) and the case-study result JSONs. At $k = 1$ the affine map equals the mean offset by construction and the local ridge degenerates to a one-label constant.

7 S7 Permutation and alignment details

Twenty uniform random permutations per A_k (identity allowed), shared between the affine and local-ridge diagnostics; losses are computed per permutation and then averaged. Real arms use calibration repeats 0–19 (identical to the permuted arms) and reproduce the main-protocol cells within 6×10^{-14}. The diagnostic is retrospective and is not a formal permutation significance test.

8 S8 Allocation sensitivity

Five frozen test-set allocations (0 = the fixed design used in all main tables; 1–4 = independent hash-seeded allocations). Main-cohort MAE moves by ≤ 0.4 pp (OCM) and ≈ 8 mg h^{-1}g$_{cat}^{-1}$ (STY) at $k = 5$; allocation 0 lies on the favourable side for OCM-F1 local ridge (reported explicitly). Method rankings are two-sided stable but not universally invariant (OCM-F1 local ridge: ranks 1–4).

9 S9 Boundary sensitivity and negative predictions

Primary predictions are never clipped. Bound sensitivity uses target-correct bounds: percent targets $[0, 100]$, STY $[0, \infty)$. The previous revision's universal $[0, 100]$ clip was an error (49.35% of STY labels exceed 100): it inflated the STY local-ridge k=10 MAE to 109.1 instead of 54.5. Negative base predictions were absent in all sampled parts.

10 S10 Suvarna 2022 reproduction details

Repository commit `0dac9383` (MIT). The notebook loads `Combined New_V1.xlsx`, which is not public; four documented compatibility fixes were applied to the published 1,234-row dataset file. Author-committed outputs (XGB test R^2 0.9132, RMSE 77.27) are the reference. Our close reproduction with per-fold train-only preprocessing: XGB 0.9210 ($+0.008$), RF 0.8806 (-0.007), GBT 0.9138 ($+0.012$). The controlled record-vs-source contrast (same rows, features, configs; only the split differs): XGB source-equal MAE 57.9→124.0, stacked R^2 0.876→0.319; nested 130.3–135.3. The strict F17-vs-F5 paired comparison (same rows/folds; F5 contains no categorical columns): record-level splits favour F17 for all three models; under source-disjoint evaluation GBT/RF advantages vanish and only XGB retains a significant advantage (-29.6 [$-42.7, -17.1$]). The earlier "F5" that contained four categorical columns is renamed F9 (reduced set) and is SI-only; the cross-dataset "buy back ≈ 4 units" comparison is withdrawn.

11 S11 Xu 2026 reproduction details

Repository commit `b7291de8`. Dataset: 1,839 rows; 82 rows without DOI are excluded from the main audit (fixed record set 1,757 rows / 156 canonical sources; 157 raw strings normalized, 1 merge). Author flow reproduced unmodified: CatBoost test R^2 0.8911 (CV 0.8887 ± 0.0121, log1p space). Leakage auto-test: five checks pass (maps/medians/scaler are functions of training rows; unseen categories $\rightarrow -1$; tampering held-out rows changes nothing). Tiers (log1p space, row-prediction-derived): B XGBoost 0.896 (random) →

0.251 (source-disjoint), CatBoost 0.881 → 0.267; For the corrected matched-five-fold comparison, source-equal MAE is 69.6198→126.3649 (XGB 3.2.0; five common seeds; pool errors within source/seed first). The original 80/20 MAE is separately 62.9776 and is not the five-fold baseline. These values supersede the old weighting claim. The archived model-configuration layers remain separate: C nested XGB 128.2 / CatBoost 129.2; D frozen X0-optimal CatBoost parameters (auto-read, hash recorded) 129.6. X5 (SI, exploratory): two-stage train-only DOI-fixed-effect residualization with `fit_intercept=True` (unit test on constant-column/intercept behaviour passes) degrades source-equal MAE to 160.8 — no benefit; the earlier concat-dummies implementation and its interpretation are withdrawn. Environment divergence: under pandas 3 the author's `dtype=='object'` check silently drops all composition columns; our per-fold pipeline detects by intent (non-numeric).

12 S12 Preprocessing leakage tests

Per-fold fitting is verified by an automated test: with held-out rows tampered (categories replaced by unseen strings, numerics set to extreme values), refitting the preprocessing on the same training rows yields bit-identical category maps and medians; unseen test categories encode to -1 and do not extend the vocabulary. The test is part of the integration suite (sixteen tests) and runs in the clean-run pipeline.

13 S13 Source registries and provenance summary

Per-corpus registries (`canonical_source_id`, `leakage_group_id`, per-row upstream file/row references) and exclusion logs ship with the reproduction package; case-study registries (`XU_SOURCE_REGISTRY.csv`, `XU_ROW_PROVENANCE.csv`, `SUVARNA_*`) record the raw-to-canonical DOI normalization (157 raw strings → 156 canonical, 1 merge) and the 82 DOI-missing rows (excluded from the main audit, sensitivity only).

14 S14 Reproducibility environment

Python 3.13.15; numpy 2.5.2; pandas 3.0.5; scikit-learn 1.9.0; xgboost 3.4.1; case-study additions: catboost 1.2.10, lightgbm, optuna 5.0.0, shap, seaborn. BLAS threads pinned to one. Cluster bootstrap: 100,000 draws, seed base 510,927, chunked. Original-author environments differ (not bit-identical); reproduction classes reflect this. All main-script logs end with exit code 0; failed earlier attempts are archived under `history/failed_runs` and marked FAILED.

Supplement B: Extended analyses
Source-disjoint evaluation in literature-derived catalysis ML

Zhang Junyu & Li Wenjie

1 Analysis populations and provenance

Historical cleaned input snapshots are inherited from the supplied research archive; the preceding full-study release refit the controlled split analysis and established a traceable source-tuned adaptation engine. It does not claim independent re-extraction of every historical measurement or restoration of the original archive's missing executed-engine version. Rows preserve original-file and row identifiers; historical OCM includes the row-audit table. The cleaned snapshots, their hashes and actual train/test ledgers are the reproducible starting point. Historical raw-file curation exclusions are not represented as newly independently audited extraction decisions.

The methanol paired corpus has 1,245 rows, 82 sources and 80 leakage groups; STY has 1,230 rows and 109 sources; OCM has 1,794 rows, 408 sources and 405 groups. Methanol and STY share 20 publication identifiers. OCM F0/F1 have identical outcome and source populations. All reported targets are finite; genuine zeros remain observations. Required operating-variable meanings follow the released upstream datasets. Missing operating values are imputed using a median fitted on the current training rows; an entirely missing training column uses the documented keep-empty-feature behavior. Numeric scaling also uses only that partition. No target-source query feature values extend composition vocabularies.

Table B1: Historical input quality, recomputed from distributed snapshots.

Panel	Feature	Missing	Minimum	Maximum
methanol	Temperature_K	0	433	673
methanol	Pressure_bar	0	0.4	80
methanol	H2_CO2_ratio	0	1	10
methanol	GHSV_L_g_h	77	0.87	240
methanol	BET_surface_area_m2_g	230	2	974.9
ocm_f0	Temperature_K	0	773	1223
ocm_f0	CH4_partial_pressure_bar	0	0.0094	2.7
ocm_f0	O2_partial_pressure_bar	0	0.003	1.35
ocm_f0	CH4_O2_ratio	0	0.8	38.31
ocm_f0	total_pressure_bar	0	0.106	4.5
ocm_f0	contact_time_s	44	0.0017	182.5
sty	Temperature_K	0	433	673
sty	Pressure_bar	0	1	360
sty	H2_CO2_ratio	0	1	20
sty	GHSV_L_g_h	0	0.9	120
sty	SBET_m2_g	0	2	1102

There are 497 missing catalyst names in the paired methanol data. The name holdout uses only nonempty normalized strings and requires a complete eligible stratum; it does not identify a physical preparation batch. The STY operating snapshot has no catalyst-name field. Richer public Suvarna metadata are retained in the retrieved workbook but are not retroactively added to this operating-input benchmark. The OCM catalyst number identifies an upstream record, not repeated measurements certified on one physical catalyst.

2 Chronology and claim scope

The historical adaptation study is retrospective: archived outcomes motivated its research question. The new-source protocol was locally timestamped and hashed before new query outcomes were evaluated, not independently preregistered. Its primary hypothesis is the five-label composition-enriched blend versus local response contrast. Bibliographic identity corrections remove re-listed old sources even when spelling or number changes; no error-dependent inclusion is used. The additional compilation is a separately released source of observations, rather than a prospective laboratory campaign.

The label-permutation, matched-alpha and fit-regime sensitivity analyses are explanatory follow-ups, not replacements for the frozen primary test. The latter uses every one of the 25 historical outer-training fits; no fit is selected by external performance. Fixed ridge/boosting split audits on old data are retrospective, with configuration held constant within each comparison. Published-workflow audits serve a separate question about validation scope. Suvarna numbers are recomputed from archived predictions; Xu's matched five-fold grouping comparison is newly refitted with XGBoost 3.2.0. Neither is external validation of the new adapters.

3 Split and support construction

Historical outer folds are stored in leakage-group split ledgers, with seeds 42, 101, 202, 303 and 404. Each seed covers every row exactly once as an outer test observation. Training inner folds randomly permute sorted leakage-group identities into three blocks. All imputation/scaling/composition encoders are fitted on the relevant inner training block before OOF prediction.

For a source with n records, sort row identifiers, shuffle with a stable SHA-derived identifier seed, and query $\max(5, \lceil n/4 \rceil)$ rows. Remaining rows are the support pool. A complete curve requires ten pool records. Fifty independently seeded pool orders create nested budgets 1, 2, 3, 5 and 10; the same order is used for every method. No support record is also a query. Calibration labels are disclosed to the prediction function only for that budget.

High-temperature design enumerates observed cutoffs and selects the upper block closest to a quarter of available finite-temperature records, subject to at least five queries and ten strictly lower-temperature pool rows. It never splits tied temperatures across support and query. Name-stratum design enumerates complete reported-name strata with the same minimum sizes and selects the query size closest to a quarter; it never places the same normalized name on both sides. Stress hyperparameters are frozen from random-source training. Matched-source random panels are available for interpretation, but their query records differ from stress panels, so these contrasts are not causal measurements of a difficulty increase.

4 All nine strategies

Let $p_A = f_G(A)$, $p_T = f_G(T)$, support responses y_A, support residual mean $\bar{r}$, and historical response scale $s > 0$.

1. Global: $q = p_T$, a zero-support-label reference at every plotted budget.
2. Support mean: $q = \bar{y}_A$.
3. Offset: $q = p_T + \bar{r}$.
4. Affine: $z_A = (p_A - \bar{p}_A)/s$, $z_T = (p_T - \bar{p}_A)/s$,

$$q = p_T + \bar{r} + z_T \frac{z_A^\top (y_A - p_A)}{z_A^\top z_A + \lambda}.$$

This shrinks the additional slope toward an offset map, with an unpenalized residual mean.

5. Empirical-Bayes offset: $q = p_T + \mu + \eta_k(\bar{r} - \mu)$,

$$\eta_k = \frac{k\tau^2}{\sigma^2 + k\tau^2}.$$

   Historical OOF residuals provide the source-equal residual mean μ, pooled within-source variance σ^2, and between-source variance corrected for within-source mean noise, truncated below at zero. The variance floor is 10^{-12}.

6. Local ridge: fit y_A on training-fitted z_A, intercept unpenalized.

7. Temperature ridge: the same support fit using only the scaled temperature coordinate.

8. Residual ridge: fit $y_A - p_A$ on z_A and add p_T.

9. Shrinkage blend: combine local and residual predictions with support leave-one-out and historical pseudo-source shrinkage, as specified in the main text.

Ridge/affine strengths are separately selected per method and label budget from $\{.1, 1, 10, 100\}$. Ties follow the declared grid order. The base grid has four ridge candidates; boosting candidates (100,2), (100,3), (300,2), (300,3), (500,8), where pairs are trees/depth. Boosting uses learning rate .05, subsample .8, minimum split size 3 and random state 42. Its candidate score is source-equal absolute OOF error, not pooled record MAE.

Adapters are scored on ten training-source pseudo-support draws per budget. Local, residual, temperature and affine strengths minimize source-equal pseudo-query MAE. For blending, prior $w_0 \in \{0, .25, .5, .75, 1\}$ minimizes pseudo-query error; disagreement scale is the source-equal mean component leave-one-out squared difference (one-label query disagreement at $k = 1$). Shrinkage $c \in \{\infty, 10, 2, .5, 0\}$ is selected with the prior/scale for each pseudo-source excluding its leakage group. Actual outer or additional-source query labels are not used. Reuse of training pseudo-queries across hyperparameter stages can affect training selection, but final outer and additional-source assessments remain separate.

5 Exact ridge leave-one-out and projection identity

Center support matrix X and responses y without penalizing the intercept. For k support records, let $K = X_c X_c^\top$, $B = (K + \alpha I)^{-1}$, and

$$\widehat{y}_T = \bar{y} + (X_T - \bar{X}) X_c^\top B(y - \bar{y}).$$

The diagonal leverage is $h_i = 1/k + (KB)_{ii}$. For $k > 1$ the exact leave-one-out response prediction is

$$\widehat{y}_{-i} = y_i - \frac{y_i - \widehat{y}_i}{1 - h_i}.$$

This uses fixed historical transformations; it does not refit the historical preprocessing on support labels. Independent mathematical verification compares predictions with scikit-learn Ridge and leave-one-out refits.

For fixed features, the map from y_A to query predictions is linear. Therefore

$$R_\alpha[y_A - p_A] = R_\alpha[y_A] - R_\alpha[p_A], \quad q_R - q_L = p_T - R_\alpha[p_A].$$

The main method chooses its two alphas separately. In the exploratory matched-alpha ablation, both use the selected local alpha without new tuning. The identity is checked numerically for all historical OCM and additional-source random draws at $k = 5$. An added global component can correct a local response or add error; linearity alone does not determine which.

6 Additional OCM source reconstruction

The public updated workbook has 4,759 rows and 533 represented publication numbers. The old bibliography has 420 unique numbers up to 421, including an ambiguous duplicated number. Existing numbers are conservatively old, even when their citation text has a typographic change. Additional numbers are screened against normalized old citations and bibliometric fields. Three aliases are removed: 424=396, 427=395 and 429=105, leaving 118 additional bibliographic entries before record-quality eligibility. Four old-number citation differences are logged; two concern missing old citation strings and the others typography. They do not qualify as new sources.

Admitted new records require finite positive temperature, partial pressures, pressure ratio and total pressure; contact time may be missing but cannot be nonpositive when observed. C_2 yield must be reported within $[0, 100]$. Composition encoding validates species/amount pairing and amount range; a fifth/sixth cation or third support cannot be omitted to fit the frozen schema. Full upstream record duplicates are removed. The exclusion reasons overlap: 1,845 records are old/unverifiable publications, 393 lack valid CH_4 partial pressure, 393 lack valid O_2 partial pressure, 100 exceed slots, 44 lack a valid target, and 21 are full duplicates. Their sum is not the excluded-row count. Exactly 2,378 distinct rows are excluded, leaving 2,381 from 91 sources.

The workbook describes species amounts as molar percentages, pressures in bar, temperature in K, contact time in s and C_2 yield as methane conversion times combined C_2 selectivity. We preserve these reported variables rather than infer pressure assumptions or physical absence from blank slots. The complete row-audit, all-source registry and bibliography screen are distributed. The 52-source main cohort and 40-source high-temperature cohort have label-blind design ledgers and shared F0/F1 support orders.

The full-fit F1 backbone is gradient boosting (500,8). At $k = 5$, local and residual alphas are both 1, affine strength 1, temperature alpha 1, blend prior .5, shrinkage 10 and disagreement scale 6.6618. F0 backbone is (100,3); local alpha 10, residual alpha 100, affine .1, temperature 1, blend prior 0, shrinkage 10 and scale 1.4613. These settings come exclusively from old OCM sources. Serialized backbones, preprocessing parameters, training OOF records and all selection scores are saved.

Table B2: Additional OCM main-cohort sources. No. is the public update bibliography number; n is the number of admitted records. Errors are pp at k=5.

No.	Citation	n	Local	Blend	Δ
422	Appl. Catal. A Gen. 219 (2001) 61-68	37	2.927	3.041	0.114
423	Appl. Catal. A Gen. 354 (2009) 143–152	60	0.570	0.753	0.183
425	Ind. Eng. Chem. Res. 48 (2009) 641–646	65	1.853	1.670	-0.183
431	J. Nat. Gas Chem. 19 (2010) 35–42	20	3.420	2.489	-0.931
434	J. Nat. Gas Chem. 19 (2010) 534–538	20	3.302	2.041	-1.260
435	J. Nat. Gas Chem. 19 (2010) 25–30	30	1.773	1.821	0.048
439	Appl. Catal. A Gen. 375 (2010) 172–178	52	2.133	2.102	-0.031
444	J. Nat. Gas Chem. 20 (2011) 204–213	45	7.283	7.467	0.184
445	J. Nat. Gas Chem. 20 (2011) 428–434	63	4.575	3.853	-0.722
446	J. Mol. Catal. A Chem. 346 (2011) 46–54	52	2.196	2.330	0.134
447	Appl. Catal. A Gen. 406 (2011) 1–12	54	4.498	4.643	0.145
448	Appl. Catal. A Gen. 391 (2011) 205–214	28	3.285	3.223	-0.063
449	J. Catal. 281 (2011) 241–253	19	4.873	4.756	-0.118
451	J. Nat. Gas Chem. 21 (2012) 49–55	27	9.719	8.658	-1.061
455	Asian Journal of Chemistry 25 (2013) 9179-9184	35	3.009	3.189	0.180
457	Appl. Catal. A Gen. 454 (2013) 100–114	50	1.466	1.250	-0.217
466	Appl. Catal. A Gen. 485 (2014) 10–19	36	4.163	4.506	0.343
467	ACS Catal. 4 (2014) 1972–1990	62	4.062	4.185	0.122
468	Appl. Catal. A Gen. 484 (2014) 97–107	25	4.732	4.446	-0.286
469	Inorg. Mater. 50 (2014) 912–916	17	3.849	4.261	0.413

No.	Citation	n	Local	Blend	Δ
470	Appl. Catal. A Gen. 472 (2014) 39–46	23	5.613	5.432	-0.182
471	Dokl. Phys. Chem. 455 (2014) 60–63	18	3.294	3.592	0.298
472	Catal. Today 228 (2014) 5–14	67	1.760	2.292	0.532
476	React. Kinet. Mech. Catal. 112 (2014) 227–240	21	4.086	4.135	0.049
480	Appl. Catal. B Environ. 150–151 (2014) 496–505	21	9.759	9.728	-0.031
481	Nanoscale 7 (2015) 2260–2264	87	5.825	5.945	0.120
482	J. Catal. 329, (2015), 560–573	26	0.432	0.479	0.048
485	ACS Catal. 5 (2015) 1663–1674	20	7.073	7.024	-0.049
486	Pet. Chem. 55 (2015) 163–168	39	3.438	3.481	0.043
492	Appl. Catal. A Gen. 528 (2016) 175–190	140	3.912	3.910	-0.002
496	RSC Adv. 6 (2016) 34872–34876	32	1.246	1.296	0.050
503	Appl. Catal. A Gen. 537 (2017) 33–39	57	1.883	1.971	0.088
504	Sci. Adv. 3, (2017)	47	3.181	3.022	-0.159
506	Appl. Catal. A Gen. 544 (2017) 77–83	48	2.532	2.556	0.024
508	Mol. Catal. 435 (2017) 13–23	19	2.221	2.316	0.095
510	ACS Comb. Sci. 19 (2017) 15–24	32	1.262	1.396	0.135
514	ChemNanoMat 4 (2018) 487–495	44	5.470	5.647	0.177
515	Appl. Catal. A Gen. 552 (2018) 117–128	31	6.306	5.840	-0.466
516	Catal. Today 299 (2018) 60–66	23	4.515	4.592	0.078
520	Chinese J. Catal. 39 (2018) 1395–1402	30	5.390	5.383	-0.007
521	Front. Chem. 6 (2018) 1–10	36	2.325	2.354	0.029
522	Eur. J. Inorg. Chem. (2018) 1787–1799	40	1.853	1.924	0.071
523	Chem. Eng. Sci. 175 (2018) 199–207	22	2.275	2.421	0.146
526	Mol. Catal. 457 (2018) 74–81	27	1.930	2.000	0.070
527	Inorg. Chem. Commun. 100 (2019) 97–100	28	5.725	5.490	-0.235
528	ACS Catal. 9 (2019) 4030–4045	18	3.406	3.711	0.305
529	Mol. Catal. 470 (2019) 40–47	146	1.595	1.796	0.201
531	J. Catal. 375 (2019) 478–492	26	4.362	4.578	0.217
535	Eur. J. Inorg. Chem. 2 (2019) 183–194	30	3.588	3.385	-0.203
536	Polyhedron 170 (2019) 602–611	63	1.200	1.175	-0.024
537	Chinese Chem. Lett. 30 (2019) 1141–1146	25	2.919	2.318	-0.601
539	Ind. Eng. Chem. Res. 58 (2019) 22847–22856	24	2.746	2.803	0.057

7 Metric estimands and uncertainty

Let e_{srj} be query MAE for source s, support repeat r and historical outer seed j. Historical source MAE is $\bar{e}_s = \frac{1}{J}\sum_j \frac{1}{R}\sum_r e_{srj}$, and the primary benchmark MAE is $\frac{1}{S}\sum_s \bar{e}_s$. External full-fit MAE omits the seed average. Row-weighted sensitivity weights each source by its fixed query count. Bounded sensitivity clips percentage predictions to $[0, 100]$ and STY predictions below at zero; raw observations are unchanged.

Cluster bootstrap samples leakage groups with replacement and divides sampled source-error sums by sampled source counts. It does not replace the source-equal estimand with group-equal averaging. Twenty thousand draws estimate percentile intervals, conditional on the fitted models and observed sources. No adjustment across secondary comparisons is applied. A zero-crossing interval is neither equivalence nor proof of no effect. Identical within-source control outputs can yield numerical differences near floating-point precision.

Stacked repeated R^2 is $1 - \sum_{sjri}(y_i - p_i)^2 / \sum_{sjri}(y_i - \bar{y})^2$ over the specified repeated prediction population. R^2 of a mean prediction instead averages each record's predictions before squaring the error; these are different quantities. The main controlled audit uses the former. An older archive field named prediction-averaged R^2 actually averaged squared errors before forming R^2; that label is not retained as an averaged-prediction metric.

8 Complete budget tables

Values below are source-equal raw MAEs. Units are pp for percent targets and mg MeOH h^{-1} g_{cat}^{-1} for STY. Stress cohorts and selected parameters are fixed across budgets. Intervals, bounded/row-weighted sensitivities and paired source values accompany the CSV exports; printed tables avoid pretending that support repeats increase the number of independent sources.

Historical / methanol / CO2_conversion_percent / random

Strategy	k=1	k=2	k=3	k=5	k=10
Global	5.271	5.271	5.271	5.271	5.271
Support mean	5.309	4.616	4.420	4.304	4.146
Offset	4.269	3.727	3.556	3.455	3.329
Affine	4.269	3.635	3.336	3.083	2.810
EB offset	4.044	3.699	3.551	3.457	3.331
Local ridge	5.309	3.633	2.914	2.429	2.039
Temperature ridge	5.309	3.526	2.905	2.519	2.267
Residual ridge	4.269	3.402	3.009	2.720	2.368
Shrinkage blend	4.315	3.394	2.870	2.441	2.040

Historical / methanol / CO2_conversion_percent / stratum_holdout

Strategy	k=1	k=2	k=3	k=5	k=10
Global	5.352	5.352	5.352	5.352	5.352
Support mean	5.832	5.112	4.878	4.673	4.581
Offset	5.081	4.494	4.301	4.155	4.080
Affine	5.081	4.608	4.333	4.167	4.028
EB offset	4.453	4.185	4.086	4.023	4.011
Local ridge	5.832	4.294	3.732	3.366	3.160
Temperature ridge	5.832	4.355	3.772	3.439	3.280
Residual ridge	5.081	4.339	4.005	3.787	3.568
Shrinkage blend	5.080	4.212	3.740	3.435	3.225

Historical / methanol / CO2_conversion_percent / temperature_high

Strategy	k=1	k=2	k=3	k=5	k=10
Global	7.536	7.536	7.536	7.536	7.536
Support mean	7.960	7.555	7.521	7.379	7.344
Offset	5.688	5.331	5.244	5.109	5.049
Affine	5.688	5.191	4.927	4.664	4.462
EB offset	5.761	5.428	5.322	5.168	5.078
Local ridge	7.960	5.455	4.098	3.210	2.834
Temperature ridge	7.960	5.262	4.099	3.477	3.067
Residual ridge	5.688	4.849	4.416	3.924	3.519
Shrinkage blend	5.818	4.933	4.129	3.275	2.849

Historical / methanol / Methanol_selectivity_percent / random

Strategy	k=1	k=2	k=3	k=5	k=10
Global	22.740	22.740	22.740	22.740	22.740
Support mean	21.217	18.676	17.803	17.136	16.680
Offset	19.098	17.130	16.425	15.747	15.395
Affine	19.098	15.873	14.155	12.539	11.502
EB offset	18.865	17.455	16.738	15.988	15.535
Local ridge	21.217	15.905	13.713	11.547	10.249
Temperature ridge	21.217	15.377	13.309	11.535	10.596
Residual ridge	19.098	15.973	14.168	12.077	10.295
Shrinkage blend	19.625	16.118	13.932	11.584	10.137

Historical / methanol / Methanol_selectivity_percent / stratum_holdout

Strategy	k=1	k=2	k=3	k=5	k=10
Global	25.729	25.729	25.729	25.729	25.729
Support mean	20.946	19.026	18.350	17.767	17.491
Offset	18.503	16.895	16.359	15.838	15.514
Affine	18.503	16.482	15.775	14.971	14.339
EB offset	19.161	17.307	16.701	16.085	15.641
Local ridge	20.946	17.205	15.962	14.510	13.911
Temperature ridge	20.946	17.512	16.092	14.900	14.455
Residual ridge	18.503	16.469	15.598	14.717	13.935
Shrinkage blend	18.799	16.496	15.527	14.523	13.809

Historical / methanol / Methanol_selectivity_percent / temperature_high

Strategy	k=1	k=2	k=3	k=5	k=10
Global	20.817	20.817	20.817	20.817	20.817
Support mean	25.201	23.391	23.329	22.853	22.609
Offset	18.291	16.696	16.482	15.992	15.608
Affine	18.291	16.116	15.544	14.170	12.659
EB offset	16.381	15.559	15.602	15.409	15.278
Local ridge	25.201	18.747	15.272	12.240	10.414
Temperature ridge	25.201	18.725	14.868	12.549	10.896
Residual ridge	18.291	15.776	14.239	12.570	10.355
Shrinkage blend	19.191	16.082	14.399	12.159	9.999

Historical / ocm_f0 / C2_yield_percent / random

Strategy	k=1	k=2	k=3	k=5	k=10
Global	5.302	5.302	5.302	5.302	5.302
Support mean	4.609	4.108	3.875	3.693	3.594
Offset	4.614	4.106	3.862	3.672	3.563
Affine	4.614	4.106	3.871	3.694	3.585
EB offset	4.527	4.164	3.947	3.759	3.618
Local ridge	4.609	4.099	3.867	3.655	3.559
Temperature ridge	4.609	4.113	3.870	3.677	3.536
Residual ridge	4.614	4.109	3.884	3.701	3.582
Shrinkage blend	4.632	4.114	3.878	3.660	3.568

Historical / ocm_f0 / C2_yield_percent / temperature_high

Strategy	k=1	k=2	k=3	k=5	k=10
Global	3.792	3.792	3.792	3.792	3.792
Support mean	3.949	3.718	3.503	3.372	3.284
Offset	4.224	3.995	3.762	3.650	3.532
Affine	4.224	3.966	3.736	3.641	3.545
EB offset	3.683	3.693	3.582	3.539	3.479
Local ridge	3.949	3.722	3.519	3.381	3.341
Temperature ridge	3.949	3.778	3.681	3.581	3.978
Residual ridge	4.224	4.011	3.799	3.691	3.711
Shrinkage blend	4.145	3.843	3.649	3.481	3.486

Historical / ocm_f1 / C2_yield_percent / random

Strategy	k=1	k=2	k=3	k=5	k=10
Global	4.527	4.527	4.527	4.527	4.527
Support mean	4.609	4.108	3.875	3.693	3.594
Offset	4.341	3.823	3.568	3.423	3.299
Affine	4.341	3.793	3.526	3.360	3.296
EB offset	3.927	3.678	3.525	3.415	3.313
Local ridge	4.609	4.080	3.710	3.374	2.839
Temperature ridge	4.609	4.113	3.870	3.677	3.536
Residual ridge	4.341	3.788	3.430	3.109	2.696
Shrinkage blend	4.331	3.773	3.408	3.081	2.633

Historical / ocm_f1 / C2_yield_percent / temperature_high

Strategy	k=1	k=2	k=3	k=5	k=10
Global	3.473	3.473	3.473	3.473	3.473
Support mean	3.949	3.718	3.503	3.372	3.284
Offset	3.449	3.086	2.957	2.899	2.807
Affine	3.449	3.080	2.941	2.882	2.708
EB offset	2.845	2.716	2.683	2.714	2.702
Local ridge	3.949	3.752	3.525	3.357	3.021
Temperature ridge	3.949	3.778	3.681	3.581	3.978
Residual ridge	3.449	3.111	2.960	2.866	2.567
Shrinkage blend	3.501	3.158	2.943	2.821	2.570

Historical / sty / STY_mgMeOH_per_h_per_gcat / random

Strategy	k=1	k=2	k=3	k=5	k=10
Global	139.258	139.258	139.258	139.258	139.258
Support mean	113.283	100.144	95.452	91.993	88.939
Offset	111.411	97.174	92.012	88.265	85.238
Affine	111.411	91.353	79.821	71.225	65.355
EB offset	104.811	94.186	90.131	86.510	84.124
Local ridge	113.283	87.602	75.645	65.210	54.901
Temperature ridge	113.283	100.188	92.917	87.098	72.872
Residual ridge	111.411	89.773	78.984	69.889	59.293
Shrinkage blend	110.732	84.321	71.439	62.608	52.771

Historical / sty / STY_mgMeOH_per_h_per_gcat / temperature_high

Strategy	k=1	k=2	k=3	k=5	k=10
Global	160.224	160.224	160.224	160.224	160.224
Support mean	154.074	144.778	140.783	139.115	136.853
Offset	139.665	130.528	126.993	124.055	121.817
Affine	139.665	132.316	122.109	113.600	106.997
EB offset	130.711	124.904	122.730	120.884	120.019
Local ridge	154.074	142.114	135.284	133.260	114.010
Temperature ridge	154.074	146.875	142.091	157.467	148.619
Residual ridge	139.665	124.610	115.905	111.808	93.301
Shrinkage blend	148.286	131.385	115.005	115.145	96.205

Additional OCM / ocm_f0 / random

Strategy	k=1	k=2	k=3	k=5	k=10
Global	5.973	5.973	5.973	5.973	5.973
Support mean	5.465	4.907	4.669	4.490	4.343
Offset	5.480	4.926	4.687	4.502	4.353
Affine	5.480	4.902	4.548	4.284	4.057
EB offset	5.201	4.891	4.710	4.545	4.381
Local ridge	5.465	4.743	4.390	4.045	3.718
Temperature ridge	5.465	4.742	4.106	3.779	3.580
Residual ridge	5.480	4.771	4.206	4.421	3.665
Shrinkage blend	5.444	4.736	4.261	4.054	3.632

Additional OCM / ocm_f0 / temperature_high

Strategy	k=1	k=2	k=3	k=5	k=10
Global	5.699	5.699	5.699	5.699	5.699
Support mean	6.204	5.886	5.712	5.556	5.436
Offset	6.217	5.843	5.674	5.512	5.384
Affine	6.217	5.813	5.484	5.209	4.946
EB offset	5.183	5.219	5.219	5.205	5.216
Local ridge	6.204	5.588	5.224	4.801	4.396
Temperature ridge	6.204	5.592	5.034	4.697	4.525
Residual ridge	6.217	5.580	5.207	5.375	4.727
Shrinkage blend	6.179	5.549	5.020	4.806	4.627

Additional OCM / ocm_f1 / random

Strategy	k=1	k=2	k=3	k=5	k=10
Global	6.120	6.120	6.120	6.120	6.120
Support mean	5.465	4.907	4.669	4.490	4.343
Offset	5.688	5.032	4.798	4.606	4.435
Affine	5.688	4.936	4.651	4.410	4.205
EB offset	5.310	4.964	4.787	4.621	4.451
Local ridge	5.465	4.492	4.022	3.593	3.252
Temperature ridge	5.465	4.742	4.106	3.779	3.580
Residual ridge	5.688	4.690	4.188	3.716	3.319
Shrinkage blend	5.428	4.469	3.996	3.552	3.230

Additional OCM / ocm_f1 / temperature_high

Strategy	k=1	k=2	k=3	k=5	k=10
Global	5.140	5.140	5.140	5.140	5.140
Support mean	6.204	5.886	5.712	5.556	5.436
Offset	6.300	5.816	5.588	5.397	5.255
Affine	6.300	5.760	5.527	5.328	5.208
EB offset	4.799	4.863	4.885	4.942	5.010
Local ridge	6.204	5.310	4.814	4.365	4.101
Temperature ridge	6.204	5.592	5.034	4.697	4.525
Residual ridge	6.300	5.307	4.861	4.454	4.233
Shrinkage blend	6.136	5.194	4.739	4.335	4.123

9 Mechanism and robustness interpretation

The external alignment diagnostic uses ten support draws, twenty permutations and five labels. Labels are permuted within the fixed support set; input records and their mean response remain unchanged. Mean, offset and EB offset therefore remain numerically invariant. The local F1 penalty is 1.2649 pp [0.8769,1.6941]; this ten-draw diagnostic is not substituted for the fifty-draw headline error. It diagnoses correspondence rather than proving chemical causality or computing a valid permutation-test p-value.

Same-alpha F1 residual differences are $-.2039$ $[-.4254, .0036]$ internally and .1230 $[-.0710, .3156]$ externally. The external alpha-matched point equals the primary residual point because both selected alphas are 1. F0 differences are .0095 internally and .0346 externally, with intervals spanning zero. The exact decomposition does not turn these inconclusive comparisons into positive evidence.

Applying all 25 old F1 outer-training fits to all additional sources at $k = 5$ gives random local/blend MAEs 3.6421/3.5797; difference $-.0624$ $[-.1603, .0244]$. High-temperature values are 4.5258/4.4357, difference $-.0901$ $[-.2066, .0167]$. These are error averages over fits, not prediction averaging, and do not replace the frozen full-fit primary result. Only three main sources report an unseen species and nine contain an operating value beyond an old input range. These coverage summaries are descriptive and do not validate a source selection rule.

10 Published-study weighting correction and reproduction scope

Archived Suvarna XGBoost predictions on 1,234 records and 109 sources yield record/source MAEs 57.8969/124.0426. The corresponding stacked raw-STY R^2 values are .8756/.3193. This release audits saved predictions; older full training-flow reproduction claims are not restated as newly executed results.

The Xu re-run uses 1,757 DOI-known rows from 156 sources; 82 DOI-missing upstream rows are outside the main audit. Its fixed XGBoost has 500 trees, depth 6, learning rate .05 and random state 42; the response is fitted in log1p space and transformed back for raw-unit MAE. Category maps and medians are fitted on each training partition. DOI is never a model input. Record and source five-fold fits use the same five seeds and current XGBoost 3.2.0. Pool record errors within each source/seed first; averaging sources directly from uneven source–fold cells produces a different weighting. Correct MAEs 69.6198/126.3649 replace the old five-fold grouping claim. Archived random-80/20 source-equal MAE is separately 62.9776. Old statements giving local OCM-F1 an average allocation advantage over affine are withdrawn; the present study's claims use its new source-tuned comparisons.

11 Reproduction files

Verified original-stage archives supply the analyses in Sections 1–11. The preceding acquisition-stage package independently contains all experiments in Section 12. The sixth release independently

contains all experiments in Section 13, including its inputs, predecessor settings and prediction exports. The current seventh release independently reproduces Section 14; earlier full releases remain separate, with exact identifiers and hashes. A final public deposit should combine these stages.

The frozen historical benchmark code is unchanged and hashes its inputs, splits and protocol. Repeated cells are preserved per fit with an index rather than duplicated in one large aggregate; the verified runner reconstructs an aggregate if requested. Historical row predictions are retained for support repeat zero, while every repeat's scalar cells remain present. All 480,500 external query prediction rows across F0/F1 include each support draw and budget. Export audit recomputes cell MAEs from those rows, checks support/query disjointness, shared representation ledgers and exact one-label identities. An atomic gzip writer is used for new large outputs; every distributed gzip stream and archive entry is validated.

The statistical engine is reproducible from the delivered cleaned snapshots. Original historical extraction, archived third-party model fitting and a prospective experimental campaign are distinct scopes. System dependency versions and commands are in the package entry-point guide. Document and plot builders consume the measured tables; no simulated performance values are used. Source attributions and third-party rights are retained separately from generated analysis code.

12 Acquisition extension: derivation, choices and complete factors

12.1 Fixed-correction risk identity

Let $G = V_T - H_A V_A$. Under the stated model, $q_A - f_T = -G\delta + H_A\epsilon$. Independence and zero means remove the cross term, yielding $\mathbb{E}\|q_A - f_T\|^2 = \operatorname{tr}(GBG^\top) + \sigma^2 \operatorname{tr}(H_A H_A^\top)$. A query-noise term, if independent and homoscedastic, adds a support-independent constant. The selector evaluates the latent-error expression. The 100,000-trial Gaussian simulation differs from the analytical risk by 0.293%. This verifies the formula under the model, not its calibration on catalyst data. Label-updated component weights make the predictor nonlinear in the support responses, so the same formula is only a design surrogate for that predictor.

The ridge map is $H_A = \mathbf{1}_T \mathbf{1}_A^\top/k + (Z_T - \mathbf{1}_T \bar{z}_A^\top) Z_{A,c}^\top (Z_{A,c} Z_{A,c}^\top + \alpha I)^{-1}$. It satisfies $H_A \mathbf{1}_A = \mathbf{1}_T$. The leave-one-out evidence has the form $D = \operatorname{diag}(1 - h_{ii})^{-1}(I - H_{AA})U_A$. Since $(I - H_{AA})\mathbf{1} = 0$, its rank is at most $k - 1$. Constants on supports have no direct independent coefficient evidence. Correlation encoded in a prior can still influence posterior predictions; this is not a claim of complete information absence in every Bayesian model.

12.2 Historical covariance construction

Only historical sources with at least 15 records enter the covariance proxy. Component scales use the source-equal first and second moments of historical cross-fitted response functions. For each source s, center $F_s = U_s/\operatorname{diag}(s_j)$ and y_s, then fit

$$\theta_s = (F_{s,c}^\top F_{s,c} + 0.1 n_s I)^{-1} F_{s,c}^\top y_{s,c}.$$

The residual proxy is $\sigma_s^2 = \|y_{s,c} - F_{s,c}\theta_s\|^2 / \max(n_s - 1 - d_s, 1)$ with ridge effective slope degrees of freedom d_s. A positive floor of 10^{-8} prevents numerical degeneracy. The within-source working posterior covariance is $V_s = \sigma_s^2 (F_{s,c}^\top F_{s,c} + 0.1 n_s I)^{-1}$. With S_θ the across-source sample covariance, set $B = 0.5 S_\theta + 0.5 \operatorname{diag}(S_\theta) + \operatorname{mean}_s V_s$ and $\sigma^2 = \operatorname{mean}_s \sigma_s^2$. These are preset empirical-Bayes proxies rather than unbiased estimates of physical noise or a proven optimal covariance.

For historical control tuning, direct covariance records from the pseudo-source's leakage group are removed. Historical cross-fitted dictionaries and inherited hyperparameters are formed within the entire outer historical-training set. This tuning stage is not a new independent test of the surrogate, and does not claim full repeated nesting of every inherited choice. The actual outer query groups and all additional-publication outcomes are excluded from all fitted covariance and control parameters.

12.3 Policies, nested budgets and query access

All selectors sort candidates by stable row identifier; exact ties choose the first. Coverage starts at the candidate nearest the pool feature centroid, then chooses a maximin point. Temperature coverage uses only standardized temperature; feature coverage uses all training-transformed features. Linear V-optimal design uses features $[10, z]$, a unit coefficient prior, and observation variance equal to the inherited local ridge α. Augmented V-optimal design uses $[10, z, U/s]$ and inherited ordinary augmented-ridge α. Its three component features are shared with the new response dictionary. Greedy V-optimal design minimizes mean query posterior variance. This is an ordinary standard-design implementation, not a claimed bitwise reproduction of Yu et al.'s complete algorithm or Fontaine et al.'s heteroscedastic online algorithm.

Innovation risk uses the exact local-ridge map in the main-text equation. Isotropic risk replaces B by $\operatorname{tr}(B)I/3$. No-component risk sets $B = 0$, leaving noise amplification only. No-noise risk sets $\sigma^2 = 0$. Pool-only risk replaces the query feature set by the complete pool. Prefix sizes $1, 2, 3, 5, 10$ are reported; intermediate sizes use the regularizer at the next reported budget. This rule maintains a common nested order. The sequence is greedy and need not minimize the final-budget global objective. V-optimal and the four query-aware risk rules see query covariates; coverage, random and pool-only risk do not. No rule sees acquisition-pool or query outcomes when selecting positions.

12.4 Predictors and historical retuning

The thirteen predictors are: global; support mean; frozen local ridge; frozen original blend; ANOVA transfer; ANOVA fixed prior; frozen local RBF; frozen ordinary two-component augmented ridge; same-local-alpha residual ridge; policy-retuned local ridge; policy-retuned blend prior; policy-retuned local RBF; and policy-retuned ordinary three-component augmented ridge. The inherited two-component features use empirical Shapley allocation with its own paired-reference intercept, rather than substituting the product-reference ANOVA functions. Their predictions remain unchanged under random acquisition to numerical precision.

Retuned local/ordinary augmented ridge choose $\alpha \in \{0.1, 1, 10, 100\}$. RBF chooses $\alpha \in \{0.01, 0.1, 1\}$ and $\gamma \in \{0.1, 1, 10\}$ with squared distances divided by the transformed feature dimension. Blend retuning chooses prior weight $0, 0.25, 0.5, 0.75, 1$, retaining its original two regularizers and shrinkage penalty. Tuning uses source-equal pseudo-query MAE, ten random orders and one order for each deterministic rule, entirely inside historical training groups. Settings are selected separately by policy and budget. All selected values and complete candidate scores are archived.

The ANOVA adapter uses inherited prior grids $\{0, 0.5, 1\}^3$ and penalties $\{\infty, 10, 2, 0.5, 0.1\}$ from the preceding response study. Its direction-aware precision is $\Omega = cD_0^2(I + D^\top D/D_0^2)^{-1}$, and box-constrained weights minimize $w^\top(D^\top D + \Omega)w - 2w^\top(D^\top e + \Omega\mu)$. Exact active-set enumeration is used in three dimensions. Infinite c returns the prior. This remains a retrospective component-transfer candidate, rather than a newly established estimator family.

12.5 Complete acquisition factors and verification

The extension contains 26 fitted regimes, 66,965 policy/budget error cells and 130 predictor–policy combinations. Nine inherited methods on random supports reproduce 510,750 error values to a maximum difference of 3.56×10^{-15}. Independent auditing re-aggregates 7,367,035 row-prediction values and re-computes 1,248,650 values directly from support labels. Every one of 13,393 support orders is reproduced after poisoning all evaluated outcomes, proving label-free acquisition for the frozen deployment interface. This does not erase retrospective corpus inspection.

Four candidate-focused random-query $k = 5$ contrasts form the reported adjusted family: innovation-risk ANOVA versus random ANOVA; versus augmented-design ANOVA; versus augmented-design retuned blend; and versus innovation-risk retuned blend. Percentile tails are $0.05/(2 \times 4)$ and its complement. Most other intervals remain nominal. Draws, seeds and folds are not independent sources. The covariance of training-fit uncertainty is not included in source-group intervals.

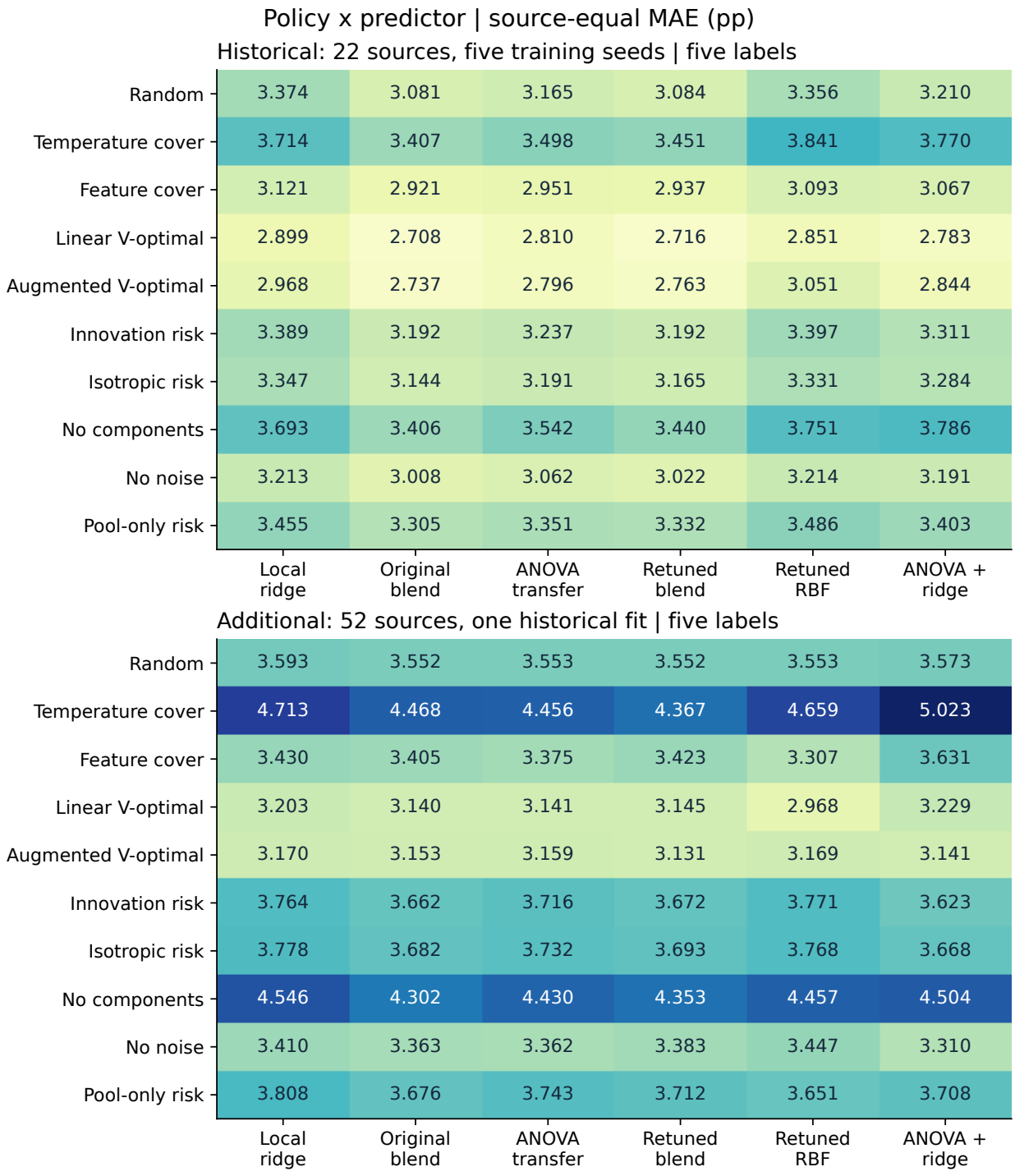


Figure B1: Selected columns of the complete acquisition factorial at five labels. The following tables retain all thirteen predictors and ten policies.

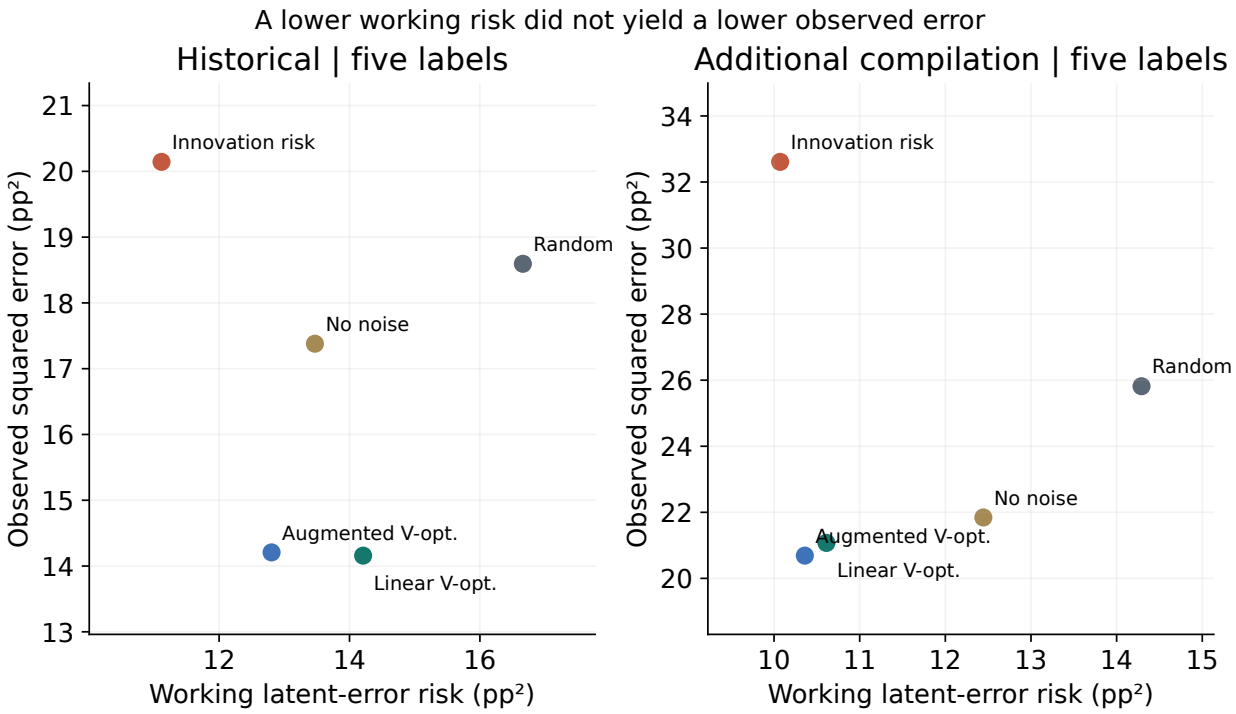


Figure B2: Secondary risk-calibration diagnostic for the fixed-prior predictor. Latent working risk excludes irreducible query noise; observed squared errors include outcome variability. Within a source, this support-independent noise cannot explain the full ordering discrepancy.

Table B3: Historical random queries, 22 sources. Five-label MAE (pp), all factors.

Predictor	R	T	F	L	A	I	S	NC	NN	P
Global	4.527	4.527	4.527	4.527	4.527	4.527	4.527	4.527	4.527	4.527
Support mean	3.693	3.935	3.491	3.409	3.497	3.672	3.666	3.947	3.709	3.775
Local ridge	3.374	3.714	3.121	2.899	2.968	3.389	3.347	3.693	3.213	3.455
Original blend	3.081	3.407	2.921	2.708	2.737	3.192	3.144	3.406	3.008	3.305
ANOVA transfer	3.165	3.498	2.951	2.810	2.796	3.237	3.191	3.542	3.062	3.351
ANOVA fixed prior	3.162	3.452	2.986	2.803	2.831	3.264	3.205	3.465	3.064	3.362
Local RBF	3.356	3.755	3.143	2.899	3.047	3.435	3.370	3.720	3.256	3.484
2-component augmented	3.127	3.521	3.028	2.718	2.736	3.239	3.190	3.524	3.067	3.313
Same-alpha residual	3.171	3.420	2.961	2.802	2.795	3.261	3.150	3.462	3.032	3.389
Retuned local	3.374	4.011	3.114	2.823	3.046	3.355	3.332	3.750	3.319	3.492
Retuned blend	3.084	3.451	2.937	2.716	2.763	3.192	3.165	3.440	3.022	3.332
Retuned RBF	3.356	3.841	3.093	2.851	3.051	3.397	3.331	3.751	3.214	3.486
3-component augmented	3.210	3.770	3.067	2.783	2.844	3.311	3.284	3.786	3.191	3.403

R: random; T: temperature cover; F: feature cover; L: linear V-optimal; A: augmented V-optimal; I: innovation risk; S: isotropic risk; NC: no components; NN: no noise; P: pool-only risk. Each column shares supports across predictors.

Table B4: Additional random queries, 52 sources. Five-label MAE (pp), all factors.

Predictor	R	T	F	L	A	I	S	NC	NN	P
Global	6.120	6.120	6.120	6.120	6.120	6.120	6.120	6.120	6.120	6.120
Support mean	4.490	5.665	4.444	4.106	4.200	4.279	4.154	5.122	4.434	4.644
Local ridge	3.593	4.713	3.430	3.203	3.170	3.764	3.778	4.546	3.410	3.808
Original blend	3.552	4.468	3.405	3.140	3.153	3.662	3.682	4.302	3.363	3.676
ANOVA transfer	3.553	4.456	3.375	3.141	3.159	3.716	3.732	4.430	3.362	3.743
ANOVA fixed prior	3.553	4.456	3.375	3.141	3.159	3.716	3.732	4.430	3.362	3.743
Local RBF	3.553	4.674	3.307	2.968	2.961	3.558	3.679	4.457	3.233	3.651
2-component augmented	3.556	4.752	3.463	3.129	3.113	3.650	3.684	4.513	3.321	3.671
Same-alpha residual	3.716	4.351	3.501	3.252	3.269	3.747	3.762	4.516	3.429	3.797
Retuned local	3.593	4.713	3.455	3.155	3.170	3.764	3.778	4.546	3.410	3.808
Retuned blend	3.552	4.367	3.423	3.145	3.131	3.672	3.693	4.353	3.383	3.712
Retuned RBF	3.553	4.659	3.307	2.968	3.169	3.771	3.768	4.457	3.447	3.651
3-component augmented	3.573	5.023	3.631	3.229	3.141	3.623	3.668	4.504	3.310	3.708

R: random; T: temperature cover; F: feature cover; L: linear V-optimal; A: augmented V-optimal; I: innovation risk; S: isotropic risk; NC: no components; NN: no noise; P: pool-only risk. Each column shares supports across predictors.

13 Source-contrast covariance transfer and matching acquisition

13.1 Why change the uncertainty model?

The earlier acquisition risk identity is mathematically correct under its stated working model, but its empirical ordering does not track observed errors well. This is motivation for a distinct candidate, not an identified diagnosis of which earlier assumption failed. The present model directly describes cross-fitted errors of γf_G after removing source levels. It combines existing random-coefficient regression, a flat intercept, functional-response features and a matching integrated-variance objective. None of empirical Bayes, universal kriging, cross-fitting or V-optimality alone is an originality claim.

13.2 Feature scaling and contrast records

For historical source s with at least 15 compatible rows, the unscaled feature vector contains the six preprocessed operating covariates and three ANOVA response functions. Each row's backbone, preprocessing and ANOVA reference set excludes that row's leakage group. Let $a_j = S^{-1}\sum_s \mathrm{mean}_i\, \phi_{sij}$ and $d_j^2 = S^{-1}\sum_s \mathrm{mean}_i\, \phi_{sij}^2 - a_j^2$, floored at 10^{-8}. Divide by d_j; an additional subtraction of a_j is unnecessary because the unrestricted source intercept makes joint feature translations irrelevant. Operating-only covariance uses the first six dimensions. All other variants use nine. These semantic coordinates allow three OOF dictionaries with different composition vocabularies to be combined without merging their full encoded feature vectors.

An outer historical fit has three first-level source-group folds. In tuning fold j, the whole held fold is absent from the covariance corpus. Within its complement, three new source-group folds supply OOF predictions for covariance rows. This requires nine secondary backbones per outer fit. Direct in-sample records from the same allowed complement form the in-sample ablation. The final deployment covariance uses the original three first-level OOF dictionaries on the complete historical training set. A fresh full historical fit reproduces the preceding full-fit geometry to numerical precision. The actual outer query groups, additional-publication outcomes and their labels never enter these fits.

13.3 Stabilized posterior-moment estimation

Define $X_s = \Phi_s - \mathbf{1}\bar{\phi}_s^\top$ and $t_s = (y_s - \gamma f_{G,s}) - \mathbf{1}\,\mathrm{mean}(y_s - \gamma f_{G,s})$. The working prior is zero-mean with covariance B. Given current B and σ^2,

$$V_s = (B^{-1} + X_s^\top X_s/\sigma^2)^{-1}, \tag{B1}$$

$$m_s = V_s X_s^\top t_s/\sigma^2, \tag{B2}$$

$$\widetilde{B} = S^{-1}\sum_s (V_s + m_s m_s^\top), \tag{B3}$$

$$\sigma^2_{\mathrm{next}} = \frac{\sum_s[\|t_s - X_s m_s\|^2 + \mathrm{tr}(X_s^\top X_s V_s)]}{\sum_s (n_s - 1)}. \tag{B4}$$

The full/operating/in-sample variants set $B_{\mathrm{next}} = 0.8\widetilde{B} + 0.2\,\mathrm{diag}(\widetilde{B})$; diagonal keeps only its diagonal, and isotropic sets $B_{\mathrm{next}} = \mathrm{tr}(\widetilde{B})I/p$. Eigendecomposition symmetrizes B and floors its eigenvalues. With $v = \max[S^{-1}\sum_s \mathrm{mean}(t_s^2), 10^{-6}]$, initialize $B = vI/p$ and $\sigma^2 = v/2$. The common positive floor is $\max(v \times 10^{-6}, 10^{-8})$. Stop after 80 updates or maximum relative covariance/noise change below 10^{-5}. Every iteration count and final change is released. The procedure is a shrunk posterior-moment fit, not a proven exact REML maximum-likelihood optimizer. Noise pools contrast degrees of freedom, whereas covariance averages posterior second moments per source. Source-equal fitting of every variance term is not claimed.

For each variant, nine pairs (γ, η) are assessed using ten random pseudo-support orders and budgets $1, 2, 3, 5, 10$. MAEs are first averaged over random orders, then equally over budgets and pseudo-sources. The minimum selects one pair for every budget and policy. The final covariance is refitted to allowed historical records at that gamma. This stage's pseudo-query scores are optimization

diagnostics. Backbone configuration and original standard-design regularizers are inherited, not independently nested again through a new global configuration search.

13.4 Established posterior and its information boundary

The main-text posterior follows the GP with an explicit constant basis and an infinitely diffuse intercept prior [17]. In a coefficient formulation, eliminating the intercept gives

$$V_A = \left[B^{-1} + \frac{\Phi_{A,c}^\top \Phi_{A,c}}{\nu}\right]^{-1}, \qquad m_A = V_A \Phi_{A,c}^\top (r_A - \overline{r}_A \mathbf{1})/\nu.$$

The predictor is $\gamma f_{G,Q} + \overline{r}_A + (\Phi_Q - \overline{\phi}_A) m_A$. For $B = \tau^2 I$ this is residual ridge with an unpenalized intercept and $\alpha = \nu/\tau^2$. Thus the isotropic version is an ordinary ridge working model; its mean transfer and its learned feature scaling are also explicit parts of the comparator.

The data information increment is $\Phi_{A,c}^\top \Phi_{A,c}/\nu$, with rank at most $\min(p, k-1)$. A component constant over supports has no independent contrast information. Off-diagonal prior covariance can couple its posterior update to observed directions, but those relationships originate in historical data, not new independent target labels. Bayesian conditioning cannot manufacture missing target information. At one label the posterior coefficient mean is zero and the prediction uses a global-response increment plus the estimated source offset.

With frozen working hyperparameters, additional support observations cannot increase latent posterior covariance in positive-semidefinite order; the flat-intercept limit is well defined for nonempty supports and positive noise. Consequently, the integrated-variance criterion is nonincreasing along the nested greedy sequence. Greedy selection need not be globally optimal, nor does decreasing working variance imply decreasing real MAE. The observation predictor's working variance adds ν to the latent diagonal. Neither variance includes uncertainty in estimated B, gamma, feature dictionary or inherited backbone choice.

13.5 Controls and statistical estimands

Policies are random, inherited linear V-optimal, inherited ANOVA-augmented V-optimal, contrast V-optimal, and contrast pool-only. Candidate pools and query rows are identical across rules. The two standard-design orders are exactly reproduced using the preceding regularizers and component scales. The contrast rule sees known query features; pool-only does not. Random is evaluated with 50 orders per source, deterministic rules once. All predictors share each policy's labels.

The eight noncandidate predictors are global, support mean, frozen local ridge, frozen original blend, re-tuned blend, re-tuned local ridge, re-tuned RBF, and ordinary ANOVA-feature ridge. Re-tuned ridge/residual controls use $\alpha \in \{0.1, 1, 10, 100\}$. RBF uses the preceding nine $(\alpha, \gamma_{\mathrm{RBF}})$ combinations. Ordinary ANOVA ridge uses inherited component scales and four alphas. Stronger blend controls first select separate local/residual alphas by source-equal pseudo-query MAE, then select a prior weight in $\{0, 0.25, 0.5, 0.75, 1\}$ and shrinkage in $\{\infty, 10, 2, 0.5, 0\}$; its evidence scale remains inherited. These controls are tuned separately per policy/budget, whereas every covariance variant keeps one random-tuned pair across all budgets/policies.

All evaluation MAEs first average orders within each source/seed, then seeds within source, then sources. Publication leakage groups are resampled in 20,000 bootstrap draws. Intervals condition on fitted models. The four locked random-query $k = 5$ comparisons are contrast-design/contrast-predictor versus linear-design/re-tuned-blend; versus augmented-design/re-tuned-blend; versus contrast-design/re-tuned-blend; and versus augmented-design/the same contrast predictor. Bonferroni percentile tails are $0.05/(2 \times 4)$ and their complement, adjusted separately within each cohort. All additional contrasts are exploratory. Prior inspection of the same corpora remains a limitation despite this within-stage protocol lock.

Support permutation is a descriptive mechanism diagnostic using repeat-zero five-label supports, twenty permutations per cell. It preserves the support mean but removes its response–input

assignment. It does not estimate prospective deployment error, identify causal catalysis mechanisms or add independent sources.

13.6 Observed performance, matched ablations and verification

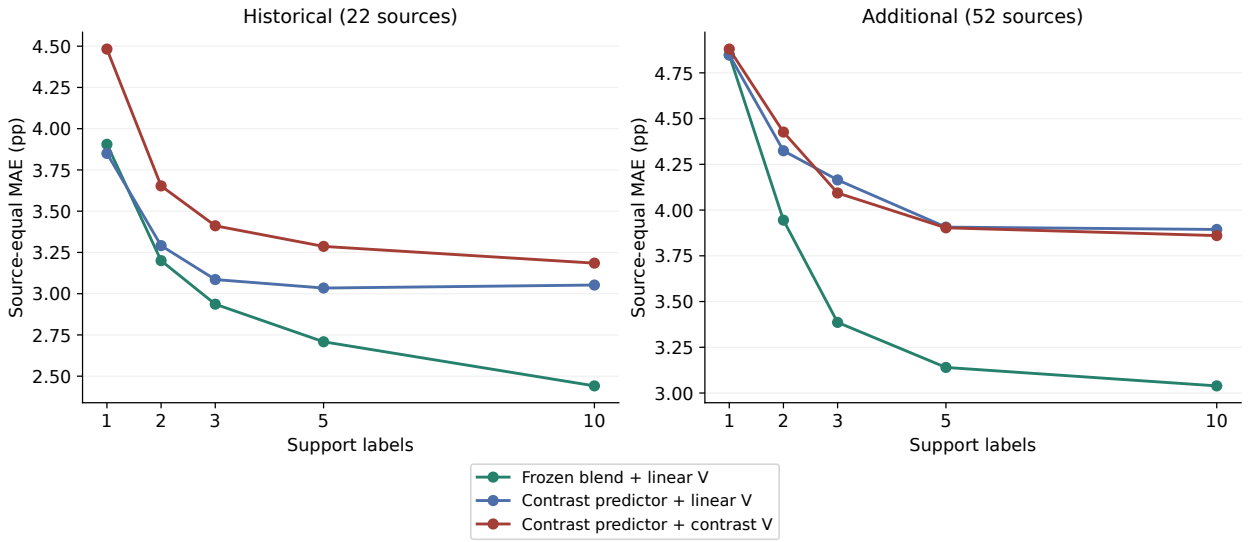


Figure B3: Random-query budgets with fixed source populations. Standard-design frozen blending remains a comparator at every budget; no curve is selected using additional-source outcomes.

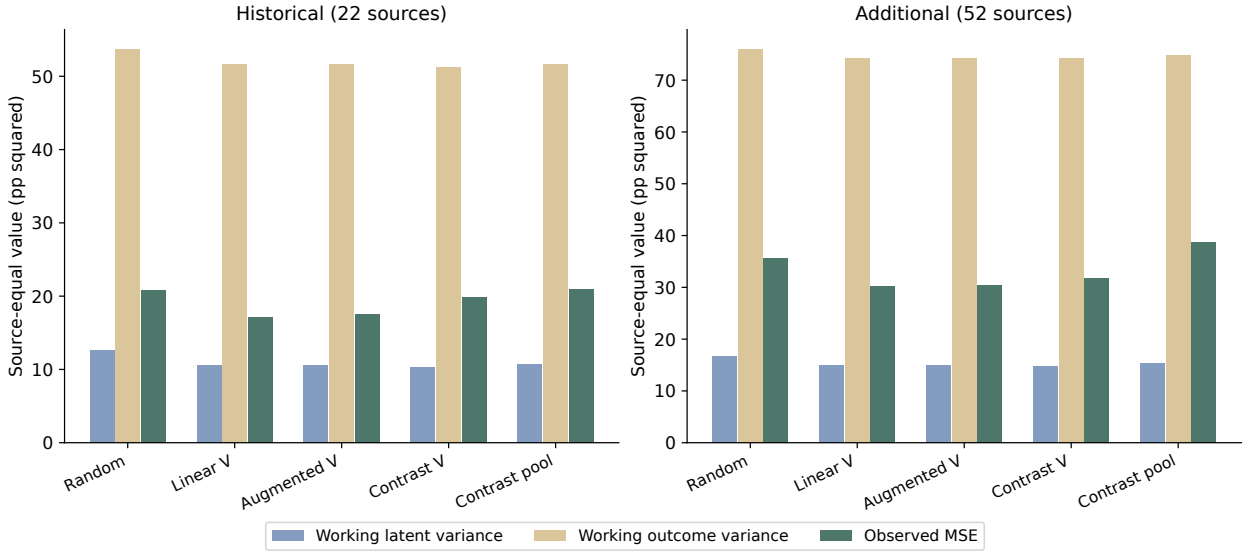


Figure B4: Five-label working variances versus observed MSE. Observation variance adds the frozen working residual term; this term includes lack of fit. These averages do not certify predictive coverage or identify physical laboratory noise.

Table B5: Historical, 22 sources. Five-label random-query MAE (pp), all 65 factors.

Predictor	R	L	A	C	P
Global	4.527	4.527	4.527	4.527	4.527
Support mean	3.693	3.409	3.497	3.624	3.675
Frozen local ridge	3.374	2.899	2.968	3.196	3.300
Frozen blend	3.081	2.708	2.737	2.920	3.030
Retuned blend	3.073	2.696	2.897	3.027	3.080
Retuned local ridge	3.374	2.823	3.046	3.304	3.297
Retuned RBF	3.356	2.851	3.051	3.303	3.222
Ordinary ANOVA ridge	3.210	2.783	2.844	3.129	3.230
Contrast full	3.365	3.034	3.090	3.287	3.389
Contrast diagonal	3.368	3.049	3.096	3.296	3.390
Operating only	3.430	3.064	3.153	3.340	3.452
Isotropic	3.390	3.059	3.107	3.300	3.413
In-sample covariance	3.382	3.065	3.063	3.291	3.369

R: random; L: linear V-optimal; A: augmented V-optimal; C: contrast V-optimal; P: contrast pool-only. All columns share query populations; each column shares supports across predictors.

Table B6: Additional, 52 sources. Five-label random-query MAE (pp), all 65 factors.

Predictor	R	L	A	C	P
Global	6.120	6.120	6.120	6.120	6.120
Support mean	4.490	4.106	4.200	4.015	4.290
Frozen local ridge	3.593	3.203	3.170	3.447	3.744
Frozen blend	3.552	3.140	3.153	3.411	3.710
Retuned blend	3.571	3.149	3.157	3.422	3.619
Retuned local ridge	3.593	3.155	3.170	3.447	3.744
Retuned RBF	3.553	2.968	3.169	3.478	3.744
Ordinary ANOVA ridge	3.573	3.229	3.141	3.349	3.709
Contrast full	4.298	3.907	4.003	3.903	4.184
Contrast diagonal	4.338	3.952	4.050	3.920	4.201
Operating only	4.405	4.032	4.114	3.981	4.280
Isotropic	3.936	3.589	3.617	3.753	3.995
In-sample covariance	4.231	3.802	3.833	3.927	4.206

R: random; L: linear V-optimal; A: augmented V-optimal; C: contrast V-optimal; P: contrast pool-only. All columns share query populations; each column shares supports across predictors.

Table B7: Post-hoc parameter-matched covariance diagnostic. Linear-design supports at five labels; gamma and total observation noise are fixed to the main candidate. Nominal intervals are exploratory.

Covariance	Historical MAE	Additional MAE
Contrast full	3.034	3.907
Contrast diagonal	3.040	3.951
Operating only	3.054	4.027
Isotropic	3.048	3.823
In-sample covariance	3.045	3.982

The candidate's historical dictionary has nine dimensions, whereas full-feature local ridge/RBF uses six operating features plus historical composition vocabulary channels. The covariance ablations are cleaner tests of covariance choices than the comparison with those wider dictionaries; neither supplies a unique causal explanation for failure. Independently tuned variants may choose different gammas/noise multipliers. The matched diagnostic was added after the locked results were observed, uses the main historical gamma and total noise, and makes no new selection from query outcomes. It evaluates all five covariance alternatives, including unfavorable ones.

The complete stage contains 61290 error cells, 65 predictor–policy combinations, and five new covariance-transfer variants. All 11,804 inherited support orders and 236,080 original-method MAEs reproduce the preceding stage to maximum 3.56×10^{-15}. The covariance stage refits 338 backbones in total: 26 full, 78 first-level and 234 secondary. The released numerical verifier checks isotropic-ridge equivalence, feature/response shift identities, all ten greedy steps against scalar scores, constant-feature nonidentifiability, and a 100,000-trial Gaussian risk simulation (0.311% relative error). Independent row audits and poisoned-outcome support-order checks are archived with exact counts. Prediction files are lossless float64 NPZ, without object arrays or quantization.

The sixth release's covariance entry point re-fits Section 13; it does not claim to newly re-fit all earlier multi-task studies. Full covariance records and arrays remain in that sixth release. The seventh release preserves the covariance summaries and figures but re-fits the separate cross-task acquisition experiment in Section 14. A separate fresh-directory smoke check re-fits one historical regime from an extracted package; the full mode re-fits all 26. Manifest verification and numerical audit use released outputs without treating a completed cache as a new training run.

14 Cross-task standard acquisition extension

14.1 Scope, chronology and fixed populations

This seventh extension evaluates five panels from four outcomes in two reaction systems, not five independent reaction datasets. Conversion and selectivity share the same paired records; twenty publication identifiers overlap the STY collection; OCM F0/F1 use the same records. These dependencies remain relevant when interpreting breadth. The current protocol is timestamped and hashed before current outer evaluation, but all input corpora were already inspected in earlier development. All new results are retrospective. Only the five-label frozen-blend query-V-minus-random comparison across the five random-query panels is primary. The parameter-matched OCM geometry crossing was added after primary results and is explicitly exploratory.

Every outer seed/fold and backbone configuration is inherited verbatim from the preceding full-study benchmark. Current outer models are re-fitted and compared with the original zero-label predictions. Historical source-disjoint OOF responses are reused for policy-specific control tuning. Inner preprocessors are reconstructed using the same three historical training-group partitions; no pseudo-validation group fits its own feature preprocessor. There is no new inner backbone fit or new global-configuration search in this extension. OOF model configuration was selected within the complete outer historical-training set in the earlier benchmark; historical control scores remain optimization diagnostics, not independent test results. Outer query groups are excluded from every such choice.

Random queries, high-temperature queries and reported-stratum holdouts use the original label-blind definitions. Eligibility is unchanged: at least ten pool records, at least five query records. Pool IDs are sorted before selection. Deterministic acquisitions are evaluated once, rather than copied fifty times as artificial replicates. Random orders run fifty times. Every policy produces one nested sequence to budget ten; intermediate steps use the original local-ridge alpha of the next reported budget. Different target outcomes can have different frozen alpha schedules, even when they share feature rows.

14.2 Five acquisitions and thirteen predictors

Temperature coverage uses the first numeric feature; full-feature coverage uses all allowed standardized/imputed numeric and encoded-composition features. The first row is nearest the pool mean and later rows maximize their minimum squared distance to the selected set. This is a conventional coverage heuristic. Query and pool V-optimal acquisition use a unit linear coefficient prior on $[10, z]$ and the inherited random-design local alpha. Query V minimizes mean working posterior variance on query covariates; pool V minimizes it over all candidate-pool covariates, including candidates that may later be observed. Neither reads outcomes. The finite intercept prior used for acquisition differs from the exact unpenalized intercept used for prediction, so working variance is only a design surrogate.

Nine original predictors are frozen: global, support mean, residual offset, affine, empirical Bayes, local ridge, temperature ridge, residual ridge and shrinkage blend. Four stronger controls re-tune local ridge, residual ridge, blending and local RBF by policy/budget. Ridge alphas are $\{0.1, 1, 10, 100\}$. RBF alpha and gamma both use $\{0.01, 0.1, 1\}$ and $\{0.1, 1, 10\}$, respectively; squared distances divide by feature dimension. Kernel ridge centers its support kernel and response and restores the support mean. After local/residual alpha selection, blending evaluates five prior weights times five shrinkages, using the inherited response scale and support-only analytic LOO. Each candidate's MAE is averaged over historical support draws, then pseudo-sources. Random pseudo-supports repeat ten times; deterministic policies once. All control grids and acquired rows are common across policies. An ordinary kernel/local fit is a strong comparator; a lower point estimate is not selected using outer query outcomes.

14.3 Primary family, units and aggregation

Each error cell averages query rows, then repeated support draws are averaged within source/seed, then seeds within source, then sources receive equal weight. A leakage group may contain more than one source; cluster bootstrap retains those source weights. Twenty thousand group resamples produce nominal 95% intervals. Five-panel primary tails are 0.005 and 0.995, the approximate Bonferroni-adjusted 99% percentile interval. Adjustment accommodates endpoint dependence but does not include training-fit uncertainty or remove retrospective selection. No folds, random draws, or shuffled-label draws are counted as independent publications. STY uses its native productivity unit, while the other outcomes use percentage points; raw errors are not pooled across these units.

Table B8: Five-panel adjusted primary results. Same interpretation and units as the main primary table.

Panel	n	Random	Query V	Gain (%)	Adjusted Δ interval
Conversion	32	2.441	2.121	13.1	[-0.707, 0.068]
Selectivity	32	11.584	10.683	7.8	[-2.140, 0.615]
STY	28	62.608	50.274	19.7	[-25.965, -2.874]
OCM F0	22	3.660	3.768	-3.0	[-0.355, 0.714]
OCM F1	22	3.081	2.708	12.1	[-0.746, -0.056]

Table B9: Frozen blend across five acquisitions at five labels. R: random; T: temperature coverage; F: feature coverage; V: query V; P: pool V. STY has native productivity units; other rows are percentage points.

Panel	R	T	F	V	P
Conversion	2.441	2.647	2.172	2.121	2.417
Selectivity	11.584	13.606	11.096	10.683	10.481
STY	62.608	70.661	53.111	50.274	58.440
OCM F0	3.660	3.901	4.008	3.768	3.895
OCM F1	3.081	3.407	2.921	2.708	2.886

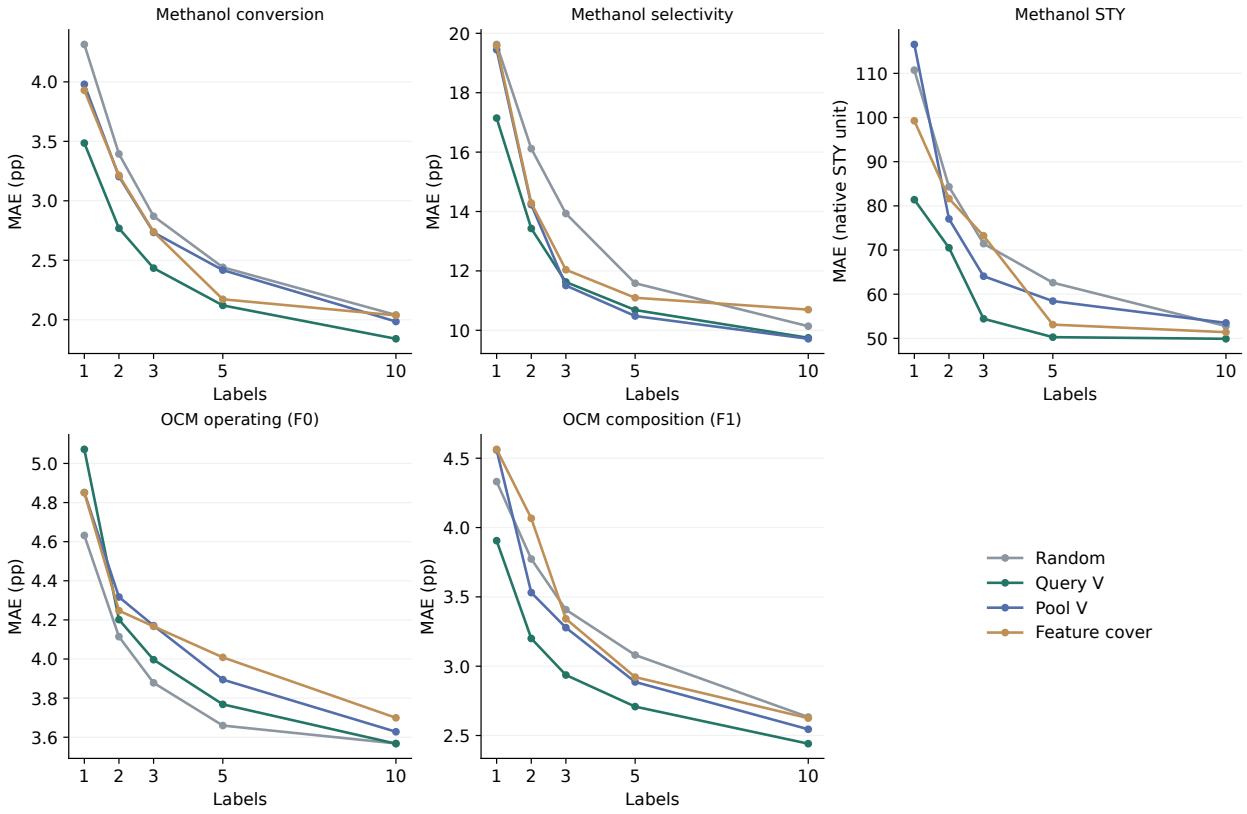


Figure B5: Budget curves for the unchanged frozen blend on five random-query panels. All rules share each panel's query population. The four displayed rules are representative; all five acquisitions and thirteen predictors are available in the numerical tables.

Table B10: Methanol conversion: all five-label random-query factors. Native STY units for STY; percentage points otherwise.

Predictor	R	T	F	V	P
Global	5.271	5.271	5.271	5.271	5.271
Support mean	4.304	4.182	4.063	3.806	4.315
Offset	3.455	3.459	3.376	3.226	3.539
Affine	3.083	3.239	2.926	2.839	3.098
Empirical Bayes	3.457	3.437	3.383	3.242	3.506
Frozen local ridge	2.429	2.576	2.197	2.135	2.394
Temperature ridge	2.519	2.711	2.470	2.343	2.587
Frozen residual ridge	2.720	2.873	2.540	2.395	2.669
Frozen blend	2.441	2.647	2.172	2.121	2.417
Retuned local ridge	2.429	2.647	2.271	2.173	2.472
Retuned residual ridge	2.720	2.856	2.524	2.421	2.668
Retuned blend	2.421	2.633	2.187	2.120	2.407
Retuned local RBF	2.401	2.531	2.075	2.153	2.436

R: random; T: temperature cover; F: feature cover; V: query V-optimal; P: pool V-optimal. Supports are common across predictors within a policy.

Table B11: Methanol selectivity: all five-label random-query factors. Native STY units for STY; percentage points otherwise.

Predictor	R	T	F	V	P
Global	22.740	22.740	22.740	22.740	22.740
Support mean	17.136	16.999	16.998	15.806	17.521
Offset	15.747	16.207	16.310	15.126	16.275
Affine	12.539	12.520	12.214	11.842	12.014
Empirical Bayes	15.988	16.439	16.481	15.424	16.501
Frozen local ridge	11.547	12.689	11.142	10.725	10.568
Temperature ridge	11.535	11.561	11.626	10.973	11.299
Frozen residual ridge	12.077	14.678	11.332	10.735	10.850
Frozen blend	11.584	13.606	11.096	10.683	10.481
Retuned local ridge	11.547	12.381	11.336	10.762	10.568
Retuned residual ridge	12.077	13.917	11.102	10.790	10.793
Retuned blend	11.623	12.159	11.097	10.691	10.579
Retuned local RBF	11.122	11.201	10.774	10.558	10.231

R: random; T: temperature cover; F: feature cover; V: query V-optimal; P: pool V-optimal. Supports are common across predictors within a policy.

Table B12: Methanol STY: all five-label random-query factors. Native STY units for STY; percentage points otherwise.

Predictor	R	T	F	V	P
Global	139.258	139.258	139.258	139.258	139.258
Support mean	91.993	110.199	101.256	77.514	97.591
Offset	88.265	104.855	92.493	78.964	90.901
Affine	71.225	82.290	69.447	62.853	71.968
Empirical Bayes	86.510	98.390	90.549	78.972	88.594
Frozen local ridge	65.210	72.805	56.486	52.671	61.580
Temperature ridge	87.098	96.672	86.843	66.248	88.206
Frozen residual ridge	69.889	76.255	62.666	57.442	65.811
Frozen blend	62.608	70.661	53.111	50.274	58.440
Retuned local ridge	65.210	69.453	56.283	51.293	60.031
Retuned residual ridge	69.889	77.153	62.784	57.352	65.539
Retuned blend	62.613	69.483	53.103	49.852	56.767
Retuned local RBF	63.957	68.185	54.535	51.547	62.927

R: random; T: temperature cover; F: feature cover; V: query V-optimal; P: pool V-optimal. Supports are common across predictors within a policy.

Table B13: OCM operating (F0): all five-label random-query factors. Native STY units for STY; percentage points otherwise.

Predictor	R	T	F	V	P
Global	5.302	5.302	5.302	5.302	5.302
Support mean	3.693	3.935	4.038	3.802	3.907
Offset	3.672	3.931	3.997	3.819	3.932
Affine	3.694	3.933	4.005	3.796	3.941
Empirical Bayes	3.759	3.963	4.000	3.866	3.956
Frozen local ridge	3.655	3.905	4.027	3.754	3.877
Temperature ridge	3.677	3.961	4.048	3.771	3.885
Frozen residual ridge	3.701	3.930	4.012	3.799	3.948
Frozen blend	3.660	3.901	4.008	3.768	3.895
Retuned local ridge	3.655	4.006	4.111	3.776	3.908
Retuned residual ridge	3.701	3.974	4.070	3.792	4.007
Retuned blend	3.665	3.989	4.072	3.774	3.945
Retuned local RBF	3.671	3.972	4.086	3.756	3.931

R: random; T: temperature cover; F: feature cover; V: query V-optimal; P: pool V-optimal. Supports are common across predictors within a policy.

Table B14: OCM composition (F1): all five-label random-query factors. Native STY units for STY; percentage points otherwise.

Predictor	R	T	F	V	P
Global	4.527	4.527	4.527	4.527	4.527
Support mean	3.693	3.935	3.491	3.409	3.688
Offset	3.423	3.682	3.195	3.041	3.429
Affine	3.360	3.564	3.193	2.976	3.365
Empirical Bayes	3.415	3.645	3.227	3.120	3.366
Frozen local ridge	3.374	3.714	3.121	2.899	3.109
Temperature ridge	3.677	3.961	3.471	3.373	3.664
Frozen residual ridge	3.109	3.332	2.940	2.759	2.927
Frozen blend	3.081	3.407	2.921	2.708	2.886
Retuned local ridge	3.374	4.011	3.114	2.823	3.114
Retuned residual ridge	3.109	3.554	3.056	2.790	2.961
Retuned blend	3.073	3.659	2.994	2.696	2.911
Retuned local RBF	3.356	3.841	3.093	2.851	3.116

R: random; T: temperature cover; F: feature cover; V: query V-optimal; P: pool V-optimal. Supports are common across predictors within a policy.

14.4 Label-correspondence diagnostic

At budget five, the first random order and each deterministic support are used for twenty outcome permutations per source/scenario/seed. The input rows, global predictions, support outcomes as a multiset, and exact support mean remain the same; only input–label correspondence changes. The frozen blend is re-computed, including its support-only blend weight. Query labels enter scoring only. These are exploratory controls, not repeated training datasets or a physical perturbation experiment. Permutation errors average shuffles and seeds before source-group intervals.

Table B15: Correspondence shuffling under query V: frozen blend, five labels. Nominal source-group intervals are exploratory.

Panel	Aligned	Shuffled	Shuffled minus aligned, 95% CI
Conversion	2.121	4.135	2.015 [1.253, 2.851]
Selectivity	10.683	16.723	6.040 [3.693, 8.513]
STY	50.274	80.945	30.671 [10.436, 59.603]
OCM F0	3.768	3.815	0.047 [-0.009, 0.134]
OCM F1	2.708	3.548	0.840 [0.478, 1.237]

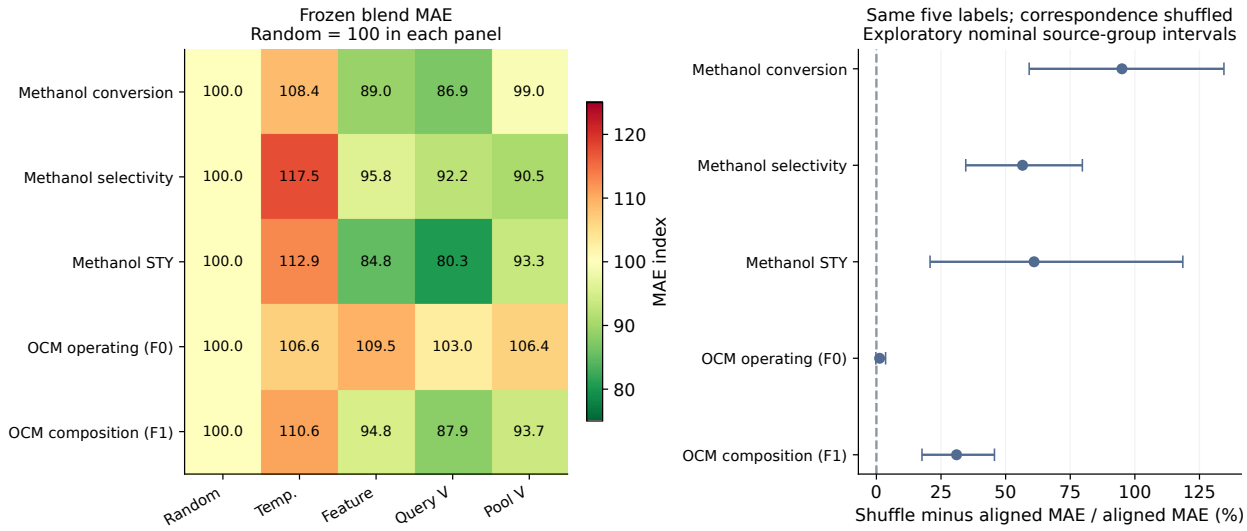


Figure B6: Left: all acquisition means for the frozen blend, normalized to each panel's observed random-design MAE. Right: outcome-correspondence shuffling effect, nominal native-unit intervals rescaled by the fixed observed aligned MAE. Normalization is for display, not a pooled or bootstrap ratio estimand.

14.5 Post-primary representation crossing

The OCM follow-up fixes the F1 alpha schedule for both operating-only (six-feature) and full composition-enriched V geometry. It evaluates each acquired sequence with both already frozen F0 and F1 predictors. Numeric preprocessors are verified equal between paired representations; outcomes, pool rows and queries are identical. The full-geometry orders exactly match the original F1 orders. The operating-only orders in this follow-up can differ from the primary F0 orders because the regularizer is now fixed to F1, rather than independently inherited from F0. A direct comparison of the primary F0 and F1 numbers would mix representation, backbone/adapter choices and acquisition regularization; this crossing removes the acquisition-alpha difference for its own conditional contrasts.

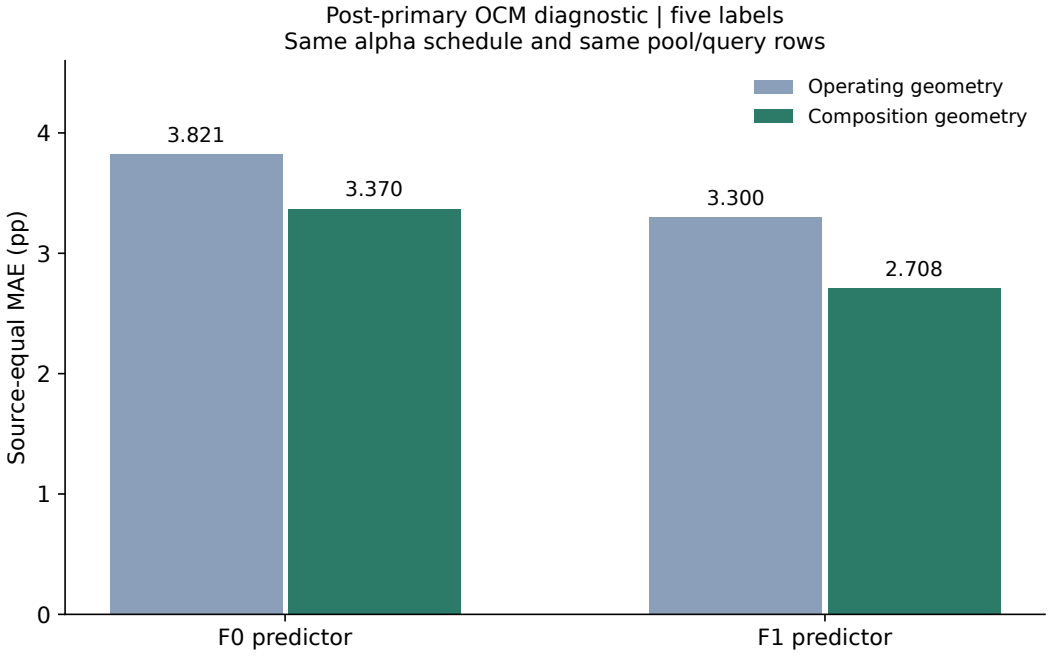


Figure B7: Post-primary exploratory crossing of selection geometry and prediction representation. Same F1 alpha, five supports, 22 OCM sources. Geometry-specific nominal intervals and all budgets are released. No new training or parameter selection occurs. Composition used for acquisition is not automatically an input to the F0 predictor.

14.6 Independent verification and reproduction scope

The current computation contains 125 outer fits, 330,750 error cells, 66,150 nested orders, and 31,712,850 scalar row predictions across 65 combinations per panel. Exactly 125 new outer backbones and zero new inner backbones are trained. Original random-design MAEs reproduce across 306,250 cells and nine methods, with maximum discrepancy 2.77e-09. The larger STY scale explains why this roundoff is not identical to the percentage-point panels. OCM frozen query/random orders and original four-method errors independently match the sixth release.

Independent audits re-aggregate 31,712,850 scalar predictions and reconstruct 2,936,375 from support rows. All 66,150 selection orders remain unchanged after replacing every target response by values between -10^9 and 10^9. Five hundred outer/inner group-separation checks pass. Mathematical verification compares all twenty greedy steps of query/pool design with a direct primal posterior and checks exact ridge prediction/LOO against independent scikit-learn fits. The post-primary crossing adds 2,700 cells and 172,800 scalar predictions, with an independent read-only audit.

The seventh release includes lossless float64 NPZ predictions, acquired/query ID ledgers, historical control scores, every parameter, input snapshots, split ledgers, predecessor references and a checksummed manifest. Its verify mode reads the release and writes audit reports only into a temporary directory. Smoke mode re-fits one OCM F1 outer fold in an empty external directory; full mode re-fits all 125 current outer models, re-tunes the current controls from inherited historical OOF records and repeats the post-primary geometry diagnostic. A completed cache is never counted as a fresh refit. The smoke check is a portability check, not a second full 125-fit independent repetition. Earlier Section 1–13 studies retain their complete raw arrays in the prior numbered releases, rather than being claimed as newly re-run by the current entry point.

15 Executed acquisition-view follow-up

15.1 Scientific scope and finite choices

This follow-up contains eleven acquisition policies and twelve predictors for each OCM prediction representation. The policies are random; operating V; composition V; feature coverage; pool V; kernel V; robust multi-view V; fixed historical best view; a zero-margin view gate; the empirical-margin conservative gate; and composition-correspondence-shuffled V. Nested budgets are 1, 2, 3, 5 and 10. Random acquisition has fifty target draws and ten historical tuning draws; chemistry shuffling has ten draws. The same support order is crossed with both F0 and F1. Gate development targets F0's frozen blend at five labels, and therefore does not claim to optimize every other predictor or budget.

The original nine response methods are joined by policy-retuned local ridge, blend and local RBF. Ridge penalties are $\{0.1, 1, 10, 100\}$. The local RBF uses the inherited nine (α, γ) combinations from $\alpha \in \{0.01, 0.1, 1\}$ and $\gamma \in \{0.1, 1, 10\}$. Blending uses the inherited finite prior-weight and penalty grid. Scores are averaged over draws and equally over historical pseudo-sources. The acquisition kernel's gamma is selected using historical F0 blend losses. The robust design is a finite greedy maximum-remaining-variance-fraction control; it is not globally optimized over all support subsets.

The operating kernel is $K_O(x, x') = 100 + x_O^T x'_O$, and composition V uses the corresponding full-feature linear kernel. Kernel V uses $K_C(x, x') = 100 + \exp(-\gamma \|x_C - x'_C\|^2 / d_C)$. For a support set A, the mean query working variance is

$$V(A) = \overline{K(q, q)} - \frac{1}{n_Q} \mathrm{tr}\{K_{QA}(K_{AA} + \alpha I)^{-1} K_{AQ}\}.$$

Robust selection minimizes $\max\{V_O(A)/V_O(\varnothing), V_C(A)/V_C(\varnothing)\}$ at each greedy step. The alpha schedule is inherited from F1 and shared by the two views. These posterior calculations are working design criteria, without a calibrated-risk assertion.

15.2 Empirical view gate and a bounded theoretical interpretation

The gate fits eight fixed response-free contexts: log pool size, log query size, log operating variation, log chemistry variation, log query chemistry novelty relative to the pool, chemistry variation unexplained by a pool-only ridge on operating features, log chemistry mean shift, and the Jaccard disagreement between the two five-point support sets. The ridge penalty is fixed at 10. Fewer than six historical groups trigger composition fallback. Otherwise group-held-out mean absolute meta errors supply the 80th-percentile margin q. Meta fitting assigns inverse within-group frequency weights. Original adapter settings had been selected on the entire outer-training set, so this margin is not fully nested calibration of every dependency.

Write $\delta = L_C - L_O$, with estimated difference $\widehat{\delta}$. A zero-margin sign choice has pointwise regret at most $|\widehat{\delta} - \delta|$ relative to the better of the two realized losses. For the empirical-margin rule O iff $\widehat{\delta} > q$, with $q \geq 0$, regret is at most $|\widehat{\delta} - \delta| + q$; the excess loss from an incorrect switch relative to staying with C is at most $\max\{|\widehat{\delta} - \delta| - q, 0\}$. These elementary identities follow directly by considering the two signs of δ. They do not control the meta prediction error under source shift, provide a coverage guarantee, or constitute a novel learning theorem. Transductive experimental design and kernel integrated-variance selection have substantial prior literature[13, 23]; the current evaluated contribution is the catalysis representation crossing and measured information decomposition.

Table B16: Locked historical primary contrasts for the conservative view gate.

Fixed control	Δ MAE	99% lower	99% upper
Composition V	0.0035	0.0000	0.0138
Operating V	-0.4484	-1.1429	0.0958
Kernel V	-0.0850	-0.2685	0.0419
Robust multi-view V	-0.0063	-0.0371	0.0147
Fixed historical best	0.0035	0.0000	0.0138

Candidate minus control; negative favors the gate. Twenty-two equally weighted sources. No contrast supports gate superiority after the five-comparison adjustment. Fixed historical best and composition V coincide here.

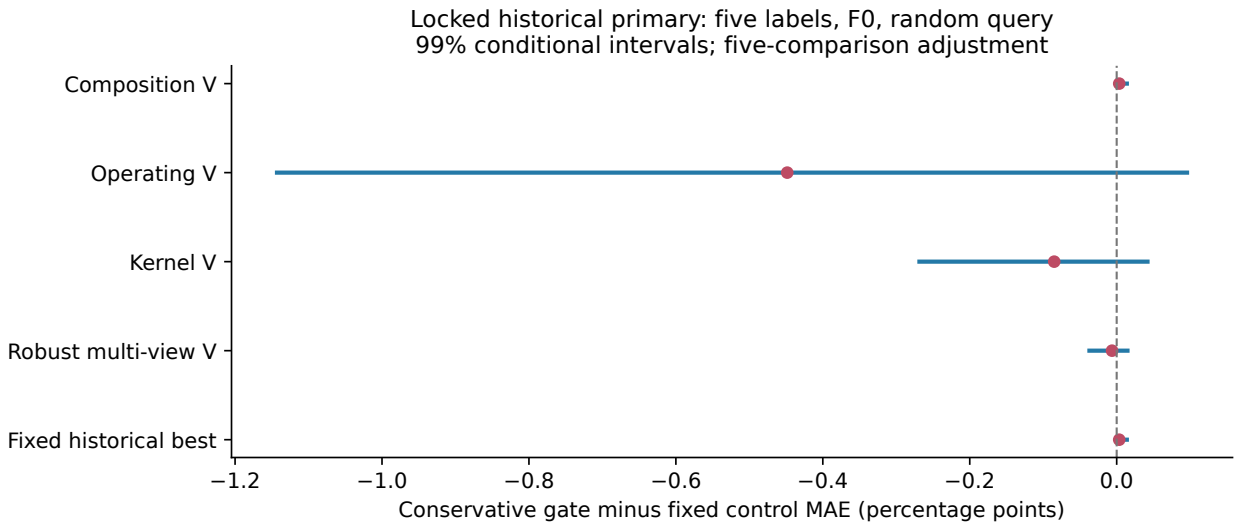


Figure B8: Historical primary gate-minus-control contrasts. Intervals are 99% conditional leakage-group bootstrap intervals for five-comparison adjustment; they average support draws and seeds before source-level resampling. No contrast supports superiority. The fixed historical best is composition V in every relevant fold, so those two controls have identical losses.

15.3 Complete acquisition results and diagnostic scope

Table B17: All eleven acquisition policies in the executed OCM follow-up.

Acquisition	Hist. F0	Hist. F1	New F0	New F1
Random	3.705	3.092	5.476	4.787
Operating V	3.821	3.300	4.817	4.483
Composition V	3.370	2.708	5.039	4.213
Feature cover	3.419	2.921	4.956	5.299
Pool V	3.657	2.886	5.175	4.015
Kernel V	3.458	2.852	5.010	4.442
Robust multi-view V	3.379	2.702	5.039	4.213
Fixed historical best	3.370	2.708	5.039	4.213
Gate without margin	3.569	2.987	5.039	4.213
Conservative view gate	3.373	2.711	5.039	4.213
Shuffled composition V	3.647	3.039	5.233	4.714

MAE in percentage points of C_2 yield. Frozen adaptive blend, five labels, random query. Hist.: 22 sources, seed-averaged; new: one publication/family, without a population confidence interval.

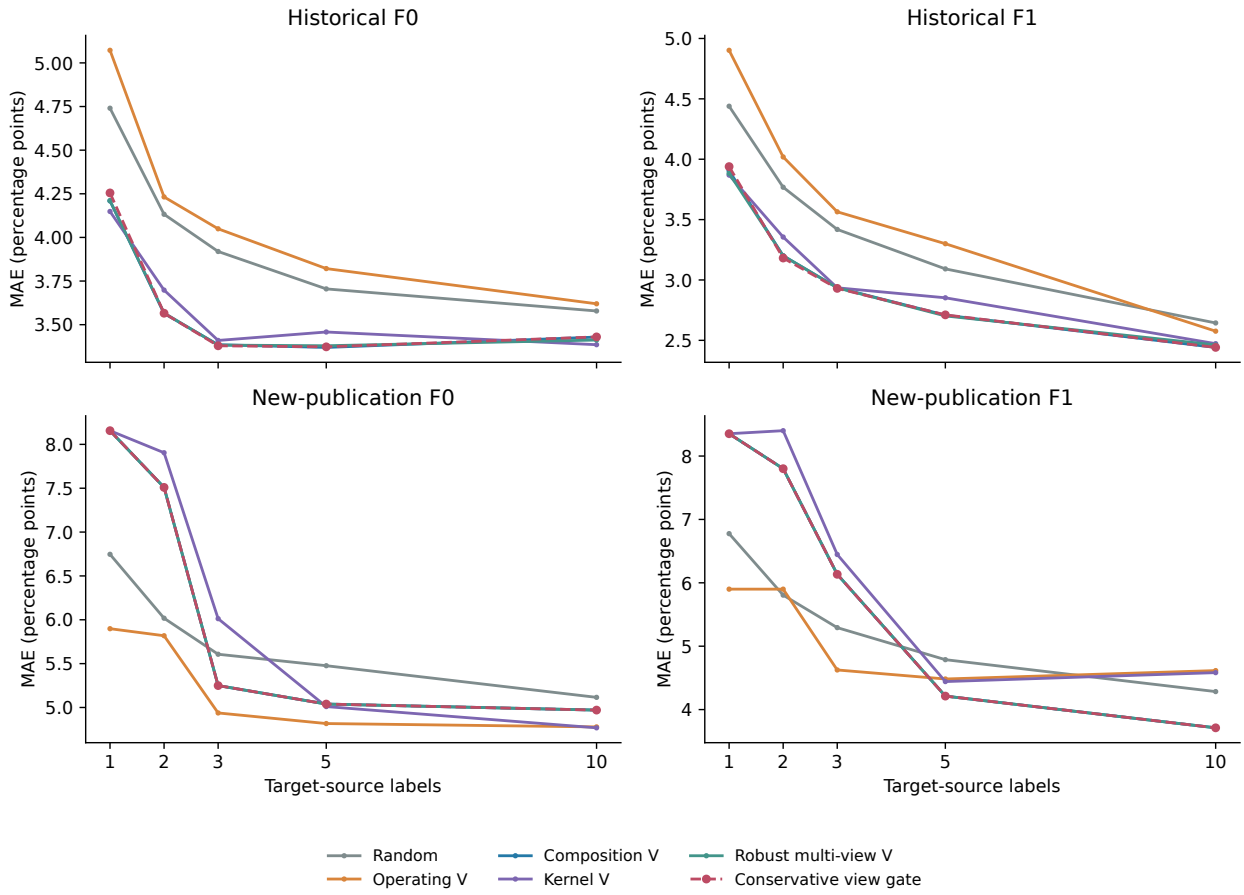


Figure B9: All five label budgets for six central acquisition policies under the fixed adaptive blend. Historical panels average sources equally; new-publication panels describe one fixed source/family. No error bars are created from support draws or from that single family. The conservative gate largely overlaps composition V. Full twelve-predictor cells, source summaries and the remaining policies are included as machine-readable results.

Electronic tables contain all twelve predictor results at five labels:

- `reports/step9_history_all_predictors_k5.csv`
- `reports/step9_new_publication_all_predictors_k5.csv`

The respective `runs/*/aggregated/summary.csv` files contain every policy, predictor, budget and evaluated scenario, including high-temperature query. Historical exploratory paired contrasts use conditional 95% intervals and are explicitly labelled as post-result diagnostics; they are not a new confirmatory family. The five-panel primary results in the preceding sections are retained unchanged. A new random seed convention explains small differences between the historical follow-up's random means and the earlier five-panel means.

16 New measurement curation and frozen application

16.1 Inclusion, units and original experiment identity

The author dataset of Nishimura et al. [22] reports measured temperature curves, validation phases, furnace length and feed ratio. Its 2928 rows include water-summary lines, repeated conditions and controls. We include only explicitly identified SVR-1, SVR-2 and Bayesian-1/2/3 validation phases, whose original study describes new catalyst preparation and testing. Mixed literature/HTS List0, proposed/predicted lists, `Unpublished`, `STD`, and `Refernce/reference` blocks are excluded. The retained population contains 1739 rows, 116 unique metal/support catalyst identities, one publication and one experiment family. The exclusion ledger records 1024 phase exclusions, 120 incomplete temperature/response entries and 45 repeated catalyst conditions. The first CSV occurrence per furnace/feed/slot-invariant composition/support/temperature signature is retained. No query responses are averaged into a support response.

Decimal commas become decimal points and Celsius becomes Kelvin by adding 273.15. Only explicitly reported feed ratios are used. The author protocol reports a 270 mm furnace, 21/7/3 mL min^{-1} $CH_4/O_2/N_2$ and 50 mg catalyst, but these facts do not supply an unambiguous gas residence volume or explicitly resolved pressure; total pressure, reactant partial pressures and contact time remain missing. The missing fractions for all four fields are 1.0. Historical-only medians are used by the frozen transformer. Precursor mmol per support mass is not converted into the historical mol-percent basis. All mol-percent fields remain missing. Single-cation oxide supports map to their support-cation symbols while original formulas are retained; oxidation state or morphology is not inferred.

Reported ethane and ethylene carbon yields sum to C_2 yield within 0.1 percentage points, and methane conversion minus reported carbon products agrees with missing carbon within 0.2 percentage points, consistent with one-decimal rounding and the original N_2 internal-standard procedure. The carbon-basis endpoint is not interpreted as an outlet mole fraction. Labels outside physical bounds would be excluded with a reason, never clipped; no such exclusion was required among eligible complete responses.

Rounded fingerprint screening finds nine old-row matches for three low-yield Na–Mn–W/SiO_2 records: 0.0% at 600/650 °C and 0.5% at 700 °C, versus old 2011 records of 0.0/0.6%. The new paper explicitly includes that catalyst among its newly prepared SVR-1 validation measurements. These nonspecific low-yield coincidences do not establish copied experimental runs; the full audit and review notes are retained. Original measurement origin, phase tracing and exclusion of mixed/reused inputs support new publication-level application. They do not certify all historical laboratory identities or full catalyst-batch independence. All phases are therefore one inference group.

16.2 Chronology, freezing and implementation verification

The twenty-five historical source/fold comparisons were completed and audited before full-history fitting. The final pipeline fits 54 group-OOF backbone candidates and two full backbones, selects

controls using historical responses only, and freezes 79 code/model/parameter/input hashes before raw candidate download and new-cohort scoring. Public paper conclusions and the CADS landing page had informed screening before this stage; the source is not a prospectively collected campaign or public preregistered test. The included row outcomes were not used to select the gate or to retune the frozen controls. Curation is locked before the first numerical application.

Historical execution exports 93150 cells, 662400 query-row records and 7948800 scalar predictions. All scalar losses are independently reaggregated, all 9315 support orders reconstructed, and 105600 query-row predictions regenerated through the prediction helpers. Actual outcome poisoning tests 2040 source/split frames, including 110 eligible frames, comparing 270 eligible partition index arrays. Separate numerical checks validate dual/primal variance equality, ridge query and exact leave-one-out values against direct scikit-learn fits, and equivalent-view robust/linear order equality. The published-source application exports 1380 cells, 620310 query-row records and 7443720 scalar values; all scalar losses and 138 support orders pass auditing and 98890 query-row predictions are reconstructed. The reconstruction reuses prediction helpers; only the loss reaggregation and direct math references are independent implementations. None of these checks is a new biological/chemical replication or a pipeline-refit bootstrap.

The ACS 2024 SI candidate was downloaded and its publisher-provided MD5 verified. Its experimental Figure S8 provides plotted replicates for three catalysts, without a complete compatible numeric row table in that asset; it is not silently digitized and counted as a second new dataset. The additional methanol candidate remains unevaluated because its publisher landing access returns HTTP403 and no compatible SI rows were acquired. These sources contribute no extra source count or numerical performance claim.

16.3 Portable reproduction and opened-cohort boundary

The new root `protocol/runtime_authorization.json` records the user's explicit replacement of the earlier private-server-only constraint. Entry points permit the authorized current runtime and keep outputs inside the project. `server/run_history.sh` performs the math check, real one-part smoke, full historical grid, audit, aggregation and final freezing. Archived Step7 bases and OOF responses are inputs to the historical extension, not claimed new outer refits. The original prepared Step8 guard/instructions are retained as predecessor provenance.

`analysis/curate_nishimura.py` writes all inclusion, provenance, missing-unit and overlap files; `code/evaluate_external_ocm.py` verifies the freeze hashes and applies the same fixed pipeline once. The manuscript sources and `analysis/build_step9_results.py` regenerate the measured figures/tables without fitting. The completed new cohort is now opened evidence. Any further gate redesign must treat it as development and obtain a different untouched publication/laboratory or a prospectively locked experiment for confirmation.

17 Predictor-matched source-local kernel development

17.1 Chronology, estimands and exact controls

This extension follows the earlier evaluated publication application. Its twenty-five historical seed/fold comparisons are method development; the same publication is subsequently used for application and stress analysis. A passed one-fold pilot had six predictors. Before the complete grid, a seventh predictor was added: the nine-grid isotropic kernel selected using exactly the same historical random support draws as the expanded kernel family. The pilot's original code and lock are preserved separately. The full-grid code and analysis lock were then fixed; no outcome-based replacement of its covariance-averaged absolute-risk primary candidate was made. All five primary comparisons and all unfavorable results are retained.

The twelve policies comprise random (50 nested draws), operating V, composition V, feature coverage, pool V, the preceding kernel V, robust multi-view V, finite-intercept GP variance,

centered squared risk, centered absolute risk, covariance-averaged squared risk, and covariance-averaged absolute risk. The seven predictors are the history-selected expanded local kernel, inherited composition-policy local RBF, frozen blend, local ridge, support mean, unadapted historical model and same-tuning isotropic kernel. The inherited RBF uses its earlier historical budget-specific parameters; these same parameters are applied across all current acquisition policies. The expanded and same-tuning isotropic kernels each use one five-label-selected parameter choice across all policies and budgets. Differences between these tuning schedules are not assigned to kernel structure.

The first seven support sequences are exactly reused from the preceding ledger, preserving paired acquisition comparisons. They are shared across F0 and F1. New support sequences are computed for each prediction representation. Within a representation, all seven predictors receive identical support rows and responses for a policy. F0/F1 differences for a new policy can include different supports and therefore are not a same-support representation ablation. The earlier seven policies supply that ablation. The fixed query inputs enter selection, so this is transductive offline design over a measured finite pool, rather than an experiment with previously unavailable query inputs.

17.2 Kernel families and historical selection

For historical-training-standardized feature vectors, define $d_B(x, x') = \|x_B - x'_B\|^2 / \max(1, |B|)$. Operating features occupy the first six coordinates, while F1's remaining coordinates describe reported composition roles, amounts and missingness. Let $K_O = \exp(-\gamma_O d_O)$ and $K_C = \exp(-\gamma_C d_C)$. The grids are

$$K_I = \exp(-\gamma d_{\text{all}}), \tag{B5}$$

$$K_{\text{add}} = (K_O + K_C)/2, \quad K_{\text{prod}} = K_O K_C, \quad K_{\text{ANOVA}} = (K_O + K_C + K_O K_C)/3. \tag{B6}$$

Every kernel has diagonal one. Nonnegative sums and products of positive semidefinite kernels give valid covariances [17]. The name ANOVA denotes this explicit sum of two block main terms and their product, not an estimated chemical interaction mechanism. Each regularizer and bandwidth is in $\{0.01, 0.1, 1\}$ and $\{0.1, 1, 10\}$, respectively. Thus F0 has $3 \times 3 = 9$ isotropic candidates; F1 has those nine plus $3 \times 3 \times 3 \times 3 = 81$ block candidates. Molar percentages missing in the opened cohort are not filled with inferred chemical compositions.

For each outer-training pseudo-source with at least fifteen records and a ten-record support pool, the inherited random query partition is held fixed. Ten deterministic random support orders each supply five labels. Its preprocessing comes from the grouped inner training fold excluding its leakage group. Candidate errors are averaged over draws, then within leakage group and equally across groups. Write the resulting error ℓ_θ. The deployed expanded kernel is $\theta^* = \arg\min_\theta \ell_\theta$. The fair isotropic control minimizes ℓ_θ over the first nine candidates with the identical draws, responses and transforms. Neither choice uses outer query or opened-publication outcomes. The complete source-wise candidate errors and chosen parameters are released. Historical outcomes had informed the research question, so this isolation does not make the full development study an unseen confirmation.

17.3 Free intercept and working risk

Kernel ridge with an unpenalized constant intercept has weights

$$A_\theta = K_{QS} C^{-1} + \frac{(\mathbf{1}_Q - K_{QS} C^{-1} \mathbf{1}_S) \mathbf{1}_S^\top C^{-1}}{\mathbf{1}_S^\top C^{-1} \mathbf{1}_S}, \qquad C = K_{SS} + \alpha I. \tag{B7}$$

To see the intercept correction, solve the constant-mean generalized least-squares estimate $\widehat{b} = (\mathbf{1}^\top C^{-1} y_S)/(\mathbf{1}^\top C^{-1} \mathbf{1})$ and substitute it into $\widehat{b}\mathbf{1}_Q + K_{QS} C^{-1}(y_S - \widehat{b}\mathbf{1}_S)$. This standard universal-kriging expression also equals the independently centered kernel-ridge solution. Summing each prediction row gives one, so replacing y_S by $y_S + c\mathbf{1}$ replaces each prediction by its original value plus c. The identity concerns constant shifts, not robustness to all domain changes.

Let a working response be $y = b\mathbf{1} + g + \epsilon$, with g of zero shape mean and covariance $\overline{K}$, and independent errors of variance ν. Since the deployed weights preserve constants, $y_Q - A_\theta y_S$ eliminates b. Its covariance is

$$\overline{K}_{QQ} + \nu I - A_\theta \overline{K}_{SQ} - \overline{K}_{QS} A_\theta^\top + A_\theta(\overline{K}_{SS} + \nu I)A_\theta^\top. \tag{B8}$$

Its diagonal is $r_q = 1 + \nu - 2(A_\theta \overline{K}_{SQ})_{qq} + \{A_\theta(\overline{K}_{SS} + \nu I)A_\theta^\top\}_{qq}$. No labels are required for this working-risk calculation. With Gaussian errors, $E|e_q| = \sqrt{2/\pi}\sqrt{r_q}$; the common constant can be dropped from selection. A nonconstant nonzero working mean, actual source bias, non-Gaussian shape or inaccurate covariance can invalidate that absolute-risk proxy for catalytic performance. Query noise is included in the scored response risk.

Centered squared and absolute risk use $\overline{K} = K_{\theta^*}$ and $\nu = \alpha_{\theta^*}$. The finite-intercept comparator instead scores mean latent posterior variance using $K_{\theta^*} + 100\mathbf{1}\mathbf{1}^\top$ and noise α_{θ^*}. Adding the same query noise constant would not change its squared-risk order. Its predictions remain the free-intercept predictor, so this comparator isolates the finite versus free intercept in the selector.

For covariance averaging, set

$$T = \max\{\text{median}_\theta|\ell_\theta - \text{median}_{\theta'}\ell_{\theta'}|, 10^{-8}\}, \quad w_\theta = \frac{\exp[-(\ell_\theta - \min \ell)/T]}{\sum_{\theta'} \exp[-(\ell_{\theta'} - \min \ell)/T]}, \tag{B9}$$

and $\overline{K} = \sum_\theta w_\theta K_\theta$, $\nu = \sum_\theta w_\theta \alpha_\theta$. These empirical weights average a working covariance and noise while leaving the deployed A_{θ^*} unchanged. Covariance averaging gives the averaged second-moment risk of these zero-mean hypotheses for a fixed estimator. Taking $\sqrt{r_q}$ instead uses the absolute risk of a covariance-averaged Gaussian model; it differs from $\sum_\theta w_\theta \sqrt{r_{q,\theta}}$ and is not the exact Gaussian-mixture absolute risk. No distributional-robustness theorem or calibrated probability interpretation is asserted.

Starting from the empty set, each policy adds the pool row giving the smallest chosen score. Pool rows are sorted by their stable identifiers; ties take the first minimum. The same ten-point order supplies prefixes of sizes 1, 2, 3, 5 and 10. Predictors use the chosen labels only after the order is computed. This implementation does not use outcome-adaptive selection, learn weights from the target labels, or optimize chemical discovery yield.

17.4 Current results and numerical checks

Table B18: Locked development primary family: averaged absolute-risk selection minus control, with the same history-selected expanded F1 kernel, five labels and random queries. Negative favors the candidate. Conditional 99% intervals adjust the five comparisons; none supports superiority.

Control	Δ MAE	99% interval	Advantage
Composition V	+0.167	[+0.001, +0.431]	No
Earlier kernel V	-0.113	[-0.437, +0.203]	No
Robust multi-view V	+0.172	[-0.002, +0.436]	No
Pool V	-0.022	[-0.438, +0.417]	No
Centered absolute risk	+0.185	[-0.009, +0.440]	No

The mathematical verification compares the free-intercept prediction against an independent centered scikit-learn kernel-ridge implementation and the inherited isotropic implementation at every budget. It checks constant-response shifts of 1000, positive semidefiniteness of all ninety F1 kernels, the working-risk diagonal against a separate joint covariance calculation, and the Gaussian expected-absolute-error identity against 50,000 simulated errors. Simulated errors are numerical checks, not catalytic measurements or empirical uncertainty coverage.

Every stored scalar query loss is independently reaggregated. All support sequences are checked for unique ten-point nesting and pool/query separation; every new selector order is reconstructed. New kernel predictions, the inherited RBF and the same-tuning isotropic predictions are checked with a separate distance/kernel implementation and centered scikit-learn fitting. Frozen blend

Table B19: All twelve acquisitions at five labels, F1: History (22 sources). MAE in percentage points. The isotropic control uses the same historical random-support tuning draws as the expanded kernel. All seven predictors and both representations are available in electronic tables.

Acquisition	Expanded	Strong RBF	Same-tuning isotropic
Random	3.317	3.288	3.378
Operating V	3.669	3.719	3.670
Composition V	2.863	2.851	2.897
Feature cover	3.085	3.056	3.142
Pool V	3.053	3.039	3.115
Earlier kernel V	3.144	3.173	3.203
Robust multi-view V	2.859	2.843	2.890
Finite-intercept GP	2.790	2.782	2.832
Centered squared risk	2.792	2.784	2.835
Centered absolute risk	2.845	2.839	2.916
Averaged squared risk	3.022	2.947	3.016
Averaged absolute risk	3.030	2.955	3.027

Table B20: All twelve acquisitions at five labels, F1: Opened source: random. MAE in percentage points. The isotropic control uses the same historical random-support tuning draws as the expanded kernel. All seven predictors and both representations are available in electronic tables.

Acquisition	Expanded	Strong RBF	Same-tuning isotropic
Random	4.848	4.792	4.792
Operating V	4.473	4.609	4.609
Composition V	4.164	3.813	3.813
Feature cover	5.219	4.801	4.801
Pool V	3.978	3.820	3.820
Earlier kernel V	4.294	4.079	4.079
Robust multi-view V	4.164	3.813	3.813
Finite-intercept GP	5.460	5.145	5.145
Centered squared risk	5.460	5.145	5.145
Centered absolute risk	5.788	5.402	5.402
Averaged squared risk	4.347	4.363	4.363
Averaged absolute risk	4.793	4.358	4.358

Table B21: All twelve acquisitions at five labels, F1: Opened source: catalyst holdout. MAE in percentage points. The isotropic control uses the same historical random-support tuning draws as the expanded kernel. All seven predictors and both representations are available in electronic tables.

Acquisition	Expanded	Strong RBF	Same-tuning isotropic
Random	4.915	4.818	4.818
Operating V	5.215	4.817	4.817
Composition V	5.055	4.680	4.680
Feature cover	5.424	4.945	4.945
Pool V	4.578	4.450	4.450
Earlier kernel V	5.033	4.562	4.562
Robust multi-view V	5.055	4.680	4.680
Finite-intercept GP	4.199	3.689	3.689
Centered squared risk	4.199	3.689	3.689
Centered absolute risk	4.199	3.689	3.689
Averaged squared risk	4.934	4.643	4.643
Averaged absolute risk	4.381	4.183	4.183

Table B22: All twelve acquisitions at five labels, F1: Opened source: high temperature. MAE in percentage points. The isotropic control uses the same historical random-support tuning draws as the expanded kernel. All seven predictors and both representations are available in electronic tables.

Acquisition	Expanded	Strong RBF	Same-tuning isotropic
Random	4.846	4.807	4.807
Operating V	5.418	5.546	5.546
Composition V	2.663	2.525	2.525
Feature cover	5.429	4.107	4.107
Pool V	4.304	5.286	5.286
Earlier kernel V	3.243	3.252	3.252
Robust multi-view V	2.663	2.525	2.525
Finite-intercept GP	2.519	2.440	2.440
Centered squared risk	2.519	2.440	2.440
Centered absolute risk	2.519	2.440	2.440
Averaged squared risk	2.501	2.437	2.437
Averaged absolute risk	2.501	2.437	2.437

Table B23: Post-hoc historical F1 contrasts at five labels with random queries. Negative favors the first named method. These nominal 95% conditional intervals have no multiplicity adjustment and do not replace the locked primary family.

Contrast	Δ MAE	Nominal 95% interval
Centered squared vs random	-0.525	[-0.797, -0.246]
Centered squared vs finite GP	+0.002	[-0.007, +0.012]
Centered absolute vs squared	+0.053	[-0.034, +0.156]
Averaged squared vs centered	+0.230	[+0.069, +0.431]
Expanded vs isotropic (composition V)	-0.033	[-0.106, +0.024]
Expanded vs strong RBF (composition V)	+0.013	[-0.070, +0.120]

Table B24: History-only selection frequency for the expanded F1 predictor in the 25 outer-training fits. Choices favor block kernels, but that frequency does not establish superiority on held-out responses. Full-history choice is additive with $\alpha = 0.01$, $\gamma_O = 0.1$, $\gamma_C = 10$.

Kernel family	Number of outer histories
isotropic	0
additive	15
product	5
anova	5

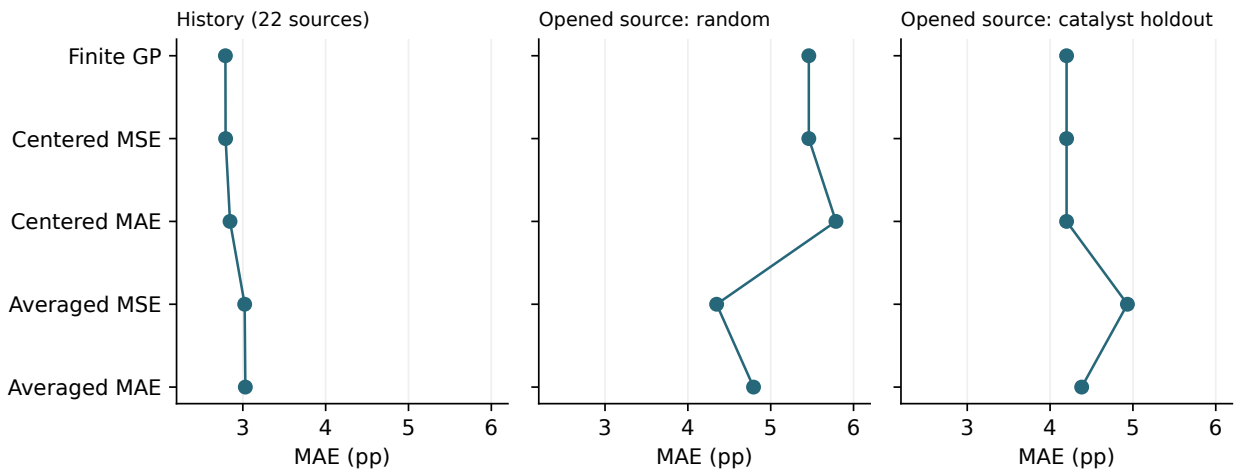


Figure B10: Five controlled changes in the working selector, evaluated with the same expanded-kernel predictor within each panel. Historical sources are equally weighted; opened-publication panels are one-family point estimates. Finite and free intercepts yield near-identical historical error, whereas covariance averaging does not establish an advantage.

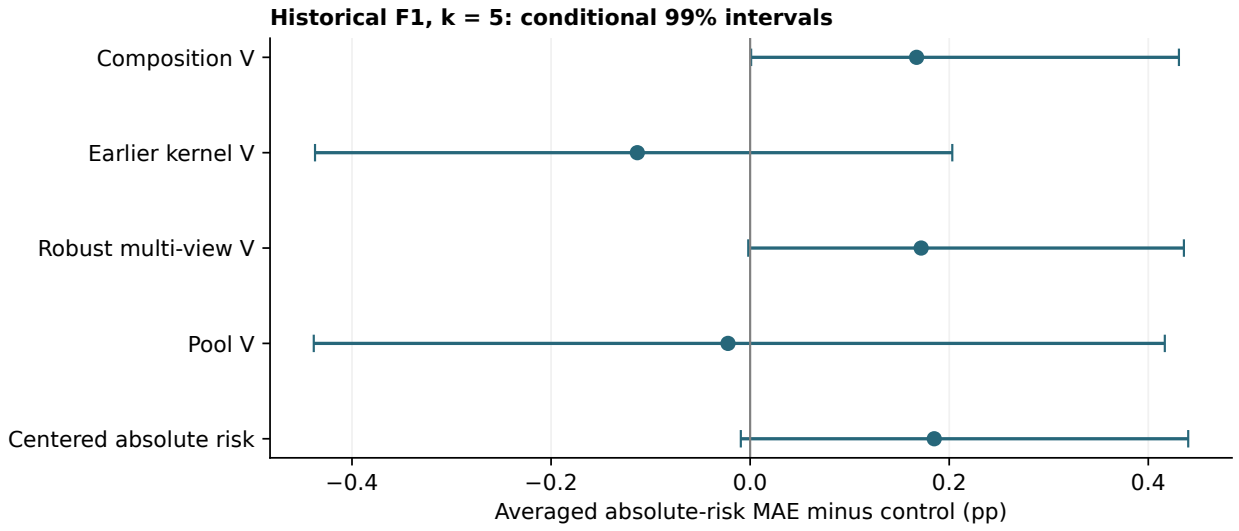


Figure B11: Locked historical development primary family: covariance-averaged absolute-risk selection minus each control under the same selected F1 kernel. Negative differences favor the candidate; conditional 99% group-bootstrap intervals adjust the five-comparison family. None supports candidate superiority.

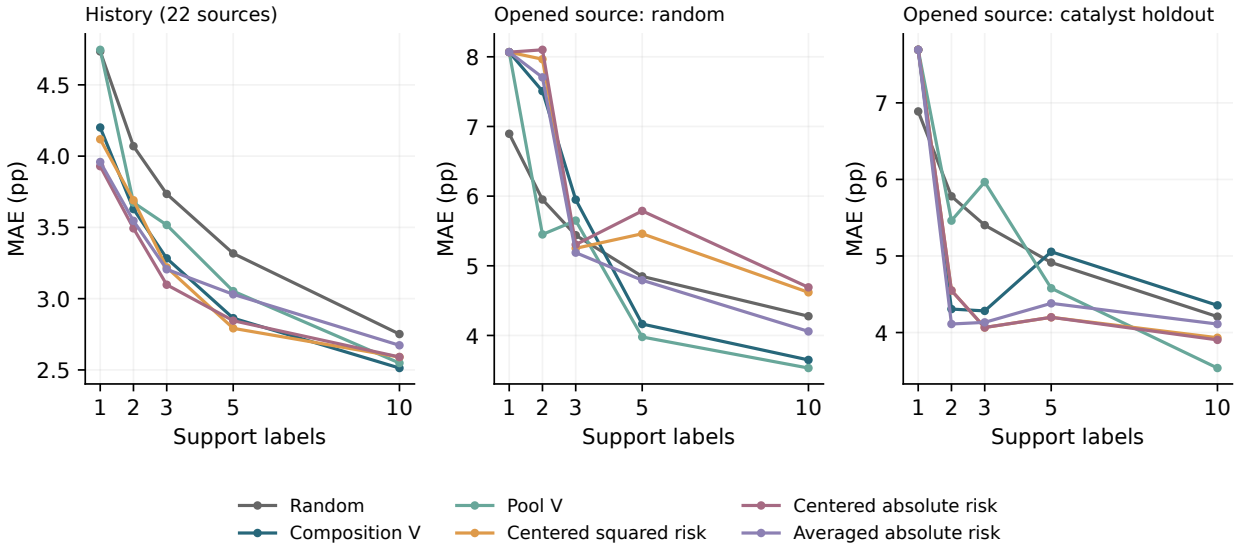


Figure B12: Budget curves for the same selected F1 kernel under six fixed acquisition policies. Catalyst holdout uses query identifiers absent from the support pool, within the already opened publication. These applications are stress/development evidence, rather than a new independent source.

reconstruction uses the inherited helpers and is identified as such. Deterministic predecessor composition-policy cells are compared exactly with the previous results. Outcome poisoning checks the inherited source partitions; the new selectors have a feature/parameter-only interface. Catalyst identifiers are separately checked for disjointness in the added stress partition. The audit does not claim that reusing a selector implementation is an independent reimplementation of the selector; the independent covariance checks provide separate mathematical evidence.

All frozen Step9 pipeline hashes, current historical-selection hashes and source ledgers are retained. This extension adds no historical or full-history global-backbone refits. Its new work is kernel candidate validation, support selection, prediction, catalyst partitioning and audit. Complete electronic tables contain all twelve policies, seven predictors, both feature representations, both historical scenarios and all three opened-publication scenarios; the displayed tables focus on five-label F1 comparisons without suppressing the remaining cells.

18 Prediction-operator covariance sensitivity

The following derivation is a fixed-design working-risk identity. It does not assert that an empirical covariance radius contains actual catalytic errors. Write the stacked observed response as $Y = (Y_S^\top, Y_Q^\top)^\top$, with mean m and covariance Σ. For a fixed response-linear predictor $\widehat{Y}_Q = AY_S$, define $B = [-A\ I]$. Its mean squared observed-query risk is

$$\mathbb{E}\|BY\|_2^2/|Q| = \operatorname{tr}(B\Sigma B^\top)/|Q| + \|Bm\|_2^2/|Q|. \tag{B10}$$

If $A\mathbf{1} = \mathbf{1}$, a common source mean $m = c\mathbf{1}$ gives $Bm = 0$. This removes a source level, not a general source-dependent mean function.

For $\Delta = \Sigma - \Sigma_0$ and $G = B^\top B \succeq 0$, a spectral bound gives

$$|\operatorname{tr}(G\Delta)| \le \|\Delta\|_2 \operatorname{tr}(G), \tag{B11}$$

$$\operatorname{tr}(G)/|Q| = 1 + \|A\|_F^2/|Q|. \tag{B12}$$

Thus the stated sensitivity bound follows for $\|\Delta\|_2 \le \varepsilon$. The upper bound is achieved by $\Delta = \varepsilon I$; $\Sigma_0 + \varepsilon I$ remains positive semidefinite. It is therefore the exact worst covariance risk over the stated

positive-semidefinite spectral ball, for the fixed prediction operator. The additive constant ε does not change which design minimizes the objective. A nonconstant mean discrepancy contributes $\|Bm\|_2^2/|Q|$ separately. Neither its size nor the covariance radius is certified empirically in this study.

The response coefficients in the main kernel equation solve centered kernel ridge with an unpenalized intercept. Their rows sum to one. Numerical checks independently construct the full joint covariance, its worst spectral perturbation and 1000 random symmetric perturbations. A 50000-draw Gaussian calculation confirms the numerical second moment within its Monte Carlo error. These are algebra checks and simulations, not measured catalysis records. All check inputs, seed and output are archived in `analysis/verify_operator_math.py` and the corresponding JSON report.

19 Additional matched-RBF and penalized-risk comparisons

Two additional acquisitions are applied with the same fixed strong isotropic RBF parameterization: nominal deployed-operator risk and nominal risk plus the amplification penalty. The RBF is the five-label strong control previously selected using historical pseudo-source composition-V supports. Its parameters are held fixed across all acquisition policies and all five budgets in this extension. At $k = 5$, this predictor equals the preceding budget-specific strong-RBF control; at other budgets, those earlier parameters can differ, so they are not identified with this fixed control.

For each of the 25 outer seed/fold fits, the four penalty values $\{0, 0.1, 1, 10\}$ are evaluated using only the allowed historical training sources as pseudo-targets. Their query/pool assignment uses a stable source/row identifier shuffle, and the acquisition functions receive transformed input arrays but no outcomes. Each candidate is scored at five support labels. Losses are averaged equally within leakage groups before candidate selection. Historical transforms exclude each pseudo-source group through the inherited grouped inner folds. This is retrospective method development on the historical program; local execution locks are not independent preregistration.

The selected penalty is zero in 23 outer fits and 0.1 in two. The full-history choice is zero. Historical random-query F1 MAE is 2.828 for nominal RBF matching and 2.913 for the selected penalized rule. The exploratory source-paired difference is +0.0855 points, with a nominal conditional 95% interval $[0.0106, 0.1850]$. The penalty therefore worsens this historical comparison on average; its exact covariance-ball interpretation does not establish predictive superiority. Its opened-publication supports coincide with the nominal RBF rule because the full-history penalty is zero. Both columns remain present so this equality is traceable.

The complete historical extension includes all twelve preceding acquisitions and the two additions, four frozen response controls, five budgets, and eligible random/high-temperature source populations. There are 42,525 new evaluation cells. The original seven-predictor factorial remains intact in the preceding computational archive; it is not reclassified as newly fitted. No global backbone is refitted in this extension. Kernel validation, penalty tuning, acquisitions and target predictions are actually evaluated.

20 Twenty catalyst-disjoint partitions

The measured-publication row-to-catalyst mapping is inherited from the curated author file. Each of twenty label-blind stable shuffles assigns 29 catalyst identifiers to queries and 87 to the support pool. All rows for an identifier stay on one side, and identity overlap is checked for every partition. Missing rows mean the query row counts can vary slightly. Query labels are used only for scoring. Random support acquisition averages fifty actual orders within each partition, with nested prefixes at budgets 1, 2, 3, 5 and 10. The same support prefix is shared by the fixed RBF, frozen blend, support mean and zero-label control. The latter control consumes no support labels.

All twenty partitions are retained. Metrics are averaged over queries and support draws within a partition, then equally over the twenty partitions. The partition table's minimum and maximum differences and its box plots describe this conditional distribution, not a sampling interval for

twenty independent publications. The original initial identity holdout and ordinary-query results are preserved separately. No best partition is selected to support the conclusion.

Table B25: Twenty additional identity-disjoint partitions at five labels, using the same fixed historical-selected RBF. MAE and differences from matched random acquisition are in percentage points.

Acquisition	Mean MAE	Mean Δ	Partition Δ range	Improved
Random	4.922	+0.000	+0.000 to +0.000	0/20
Composition V	4.107	-0.815	-1.616 to +0.120	19/20
Operating V	4.926	+0.004	-1.025 to +0.731	11/20
Pool V	4.221	-0.701	-1.272 to +0.210	18/20
Expanded-kernel risk	4.480	-0.441	-1.545 to +0.498	15/20
RBF-matched risk	4.652	-0.270	-1.360 to +0.621	13/20
Penalized RBF risk	4.652	-0.270	-1.360 to +0.621	13/20

Composition V gives mean MAE 4.107, compared with 4.922 for random acquisition, a 16.6% ratio-of-means reduction. It improves 19 of twenty partitions. Pool V improves 18, while expanded-kernel risk improves 15 and RBF-matched risk improves 13. The stable positive finding is therefore the value of chemistry-directed measurement coverage under identity exclusion; the earlier single-partition ranking does not generalize uniformly. These repeated partitions remain a sensitivity within one opened experimental family.

The historical nominal risk and actual MSE rankings are also compared within sources at five labels; this removes arbitrary cross-source response scales from a pooled rank analysis. Their median source-level Spearman correlation is 0.514. This descriptive diagnostic does not calibrate the risk in outcome units or validate the covariance radius. The full per-source diagnostic values are in the electronic tables.

21 Independent verification and reproducibility

The new computations produce 10,948,000 scalar query predictions: 1209600 across 25 historical parts and 9738400 in the opened-publication sensitivity. All scalar losses are reaggregated independently. Every one of the 2737000 fixed-RBF query values is reconstructed using independently generated Euclidean distances, scikit-learn RBF kernels, `KernelCenterer`, and `KernelRidge`. Maximum absolute reconstruction differences are below 10^{-9}; detailed tolerances and the actual maxima are reported in the archived audit JSON. Support/query row disjointness, ten unique support positions and every query ordering are checked. Source repetitions and these counts do not increase the inferential sample size.

`code/run_extension.py` computes the extension; `analysis/audit_extension.py` independently verifies predictions and losses; `analysis/build_results.py` produces presentation tables and figures without recomputing scientific predictions. All original input data, source/group allocations, fitted objects, prior complete results and code are preserved under `reference_step10/`. The release includes dependency versions and a single reproduction entry point. Reproduction does not require a private server or a GPU. Saved fitted objects require a compatible scikit-learn environment.

22 Investigation of additional measurement resources

The public CADS table linked to Nguyen *et al.*, ACS Catalysis 2021, DOI 10.1021/acscatal.0c04629, contains 300 catalyst entries, of which 291 have complete reported response/condition fields. Related author documentation identifies this table as best-yield conditions selected across multiple reaction settings. Such response-selected operating conditions change the estimand and do not supply a directly comparable fixed-condition few-label application. The dataset also explicitly restricts redistribution. It is not evaluated as an additional confirmation dataset and its raw file is not included in this release.

The open methanol kinetics study DOI 10.1039/d5re00330j describes a 500-point measured dataset. Attempts to retrieve a compatible SI table failed in this runtime. No values were inferred from figures or fabricated. The 2025 OCM author study DOI 10.1080/14686996.2025.2454219 was inspected through its open full-text XML, but its supplementary CSV was not successfully retrieved and validated. It is likewise excluded. These investigations add no new independent measured experiment family to the reported evaluation.

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